

Overlapping Calendars: Origins and Consequences of Conflicting Timing Institutions

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October 6, 2026

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Abstract

We study the origins and economic consequences of conflicts between timing institutions—rules that set common schedules—and local economic activity. We focus on school–harvest conflicts by assembling a global dataset of countries’ school calendars and comparing them with the timing of harvesting. We document systematic calendar overlap, particularly in southern-latitude countries and former colonies. Instrumenting overlap with colonial variation in calendar timing, we find that a one-percentage-point increase in school–harvest overlap raises school dropout by 0.6–0.8 percentage points. Beyond colonial inheritance, similar timing conflicts can arise through domestic standardization when a uniform calendar is imposed despite local crop variation. We study this distinct channel using a Colombian reform that shifted public schools to a common national calendar, increasing overlap in some schools while reducing it in others. Using administrative census data covering the universe of rural public schools, we find that a one-percentage-point increase in school–harvest overlap raises the dropout rate by about 0.7 percentage points, while outcomes improve where overlap falls. Our estimates are robust across alternative estimators, control groups, clustering levels, and treatment definitions. Employing nationally representative Colombian household survey data, we find agricultural work among children rises in affected municipalities, consistent with a school–labor time-allocation mechanism. These findings identify institutional timing as an important determinant of human capital accumulation in agrarian economies.

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Keywords: Seasonality, School calendar, Dropout, Colonial legacies, Colombia, Cross-Country
JEL codes: I25, J22, J24, O12, O13, O43.

1. Introduction

A large literature establishes that institutions are key determinants of long-run economic performance (North, 1991; Acemoglu et al., 2001). Most of this literature studies *what* institutions do. Property rights, legal systems, and political authority have received sustained attention as drivers of economic outcomes. However, institutions also shape behavior through a second dimension that remains largely unexplored. They determine *when* economic activity takes place.

Timing institutions are everywhere. School calendars dictate when children attend class. Office and business hours shape when adults work. Tax cycles and electoral schedules govern when major civic obligations occur. Each of these arrangements coordinates millions of people around a common temporal structure. When imposed timing aligns with the rhythm of local economic activity, they go unnoticed. When it does not, they generate frictions with relevant consequences for individual decisions and aggregate outcomes. Despite how common they are, the economic effects of timing institutions remain understudied. This paper provides systematic evidence that timing institutions are quantitatively important for economic development. We focus on a particularly consequential one, the school calendar, and study how its misalignment with local agricultural cycles shapes human capital accumulation in agrarian economies.

To assess the origins and scope of this institutional misalignment, we begin with cross-country evidence. We focus on harvest periods because they represent a particularly labor-intensive and time-sensitive phase of the agricultural cycle. Resource-constrained rural households rely more heavily on family labor during these periods, including children.¹ Schooling that coincides with the harvest creates a particularly salient competition for children’s time (Ito and Shonchoy, 2026). We, therefore, assemble a global dataset of school calendars and compare them with the harvest timing of each country’s major crop.

We find substantial school–harvest overlap, particularly in southern-latitude countries and

¹Harvest-related work can extend beyond crop collection to activities such as gathering, carrying, sorting, processing, and other pre- and post-harvest tasks, along with livestock care, making conflicts with schooling especially consequential.

former colonies. This pattern is systematically related to colonial inheritance. Many former colonies retain calendar structures that resemble those of their colonizers even when local seasonal conditions differ substantially. School–harvest overlap rises sharply as the local harvest falls farther from the colonizer’s school vacation period. Such gradient is present only among former colonies whose school calendars remain closest to those of their colonizer. By contrast, this relationship is inexistent among countries that developed less similar calendars. These patterns are consistent with persistent institutional transfer (Acemoglu et al., 2001) and with evidence that colonial education leaves durable legacies (Bolt and Bezemer, 2009). More broadly, they suggest that school calendars can remain shaped by historical institutions even when the schedule is inadequately aligned with local economic conditions.

We then use this colonial persistence to identify the causal effect of school–harvest overlap on schooling outcomes. We instrument calendar overlap using the interaction between the colonizer’s school calendar and harvest timing predicted by the former colony’s geographic conditions. Our estimates show that a one-percentage-point increase in school–harvest overlap raises the out-of-school share of primary-school-age children by roughly 0.6 to 0.8 percentage points. Our estimates remain strongly consistent when the identification depends strictly on variation among former colonies. These estimates provide evidence that inherited timing institutions can have substantial educational consequences.

This cross-country evidence is nonetheless aggregated by design, and such aggregation limits the depth of the analysis. National averages can conceal important within-country variation in crop cycles, school calendars, and schooling decisions, while country-level measures of both harvest timing and school schedules may also be measured with error when they vary across regions. Moreover, the cross-country exercise speaks to only one source of misalignment—the persistence of colonial institutions—and leaves open whether comparable conflicts can arise from other forms of institutional design. One such source is domestic standardization. A nationally uniform calendar can improve coordination across the education system, while it may remain insufficiently aligned with local agricultural conditions in regions with different crop cycles.

We study this second source of school–harvest conflict using a natural experiment in Colombia, which provides a useful within-country setting for examining the relationship between school calendars and local agricultural seasonality. This setting allows us to hold country-level institutions constant and exploit granular school-level data to examine the consequences and underlying mechanism of calendar misalignment, complementing the cross-country evidence.

Beginning in 2011, public schools in Nariño and Valle del Cauca, the only two departments in Colombia operating a September–June calendar, shifted to the January–November calendar used in the rest of the country.² Because harvest timing varies with local crop composition, the same calendar reform altered school–harvest overlap differently across locations: overlap increased in some schools, declined in others, and remained essentially unchanged elsewhere. Averaging across all affected schools, overlap increased in 60.4 percent of affected schools, was unchanged in 9.2 percent, and declined in 30.5 percent. This variation allows us to identify the consequences of both worsening and improving calendar alignment within a common institutional environment. Importantly, the reform changed the timing of instruction without altering curricula, assessments, or school resources.

Using administrative school census data covering the universe of rural public schools in Colombia, we find that a one-percentage-point increase in school–harvest overlap raises the dropout rate by about 0.7 percentage points and lowers the grade-promotion rate by 0.69 percentage points. Among schools where overlap increased, dropout increases by 0.53 percentage points per point of overlap. Among schools where overlap declined, it falls by 1.03 percentage points. These estimates correspond to about a 17 percent increase and a 23 percent decline in the respective pre-reform dropout rates. The effects persist over time and are concentrated among primary-school-age children in rural areas, the age group most likely to be engaged in family subsistence farming (ILO and UNICEF, 2025).

Our estimates are robust along several dimensions. They are stable across heterogeneity-robust estimators (de Chaisemartin et al., 2024), across control groups ranging from the full country to

²Departments are Colombia’s first-level administrative divisions.

local border samples around affected departments' boundaries, across clustering at the school and municipality level, and across alternative definitions of the treatment variable. Synthetic difference-in-differences estimates, which recover the reform's total effect on treated departments rather than the overlap channel alone, also deliver consistent magnitudes.

To understand the mechanism, we use nationally representative household survey data and a difference-in-differences design to estimate the effect of the reform on children's agricultural work. We find that the probability of children aged 10 to 15 engaging in agricultural work rose by about 4 percentage points in affected municipalities following the reform, more than half the pre-reform rate. The increase is concentrated among children who are not enrolled in school, but also present among those who remain enrolled. Taken together, the results are consistent with a time-allocation mechanism in which greater overlap between the school year and the local harvest raises the opportunity cost of schooling.

The estimated magnitudes from the cross-country and Colombia analyses are expressed in comparable units and are strikingly similar. These estimates, however, do not identify the same formal parameter. The cross-country outcome is a national stock measure of children who are out of school, identified from colonial variation in inherited calendars, whereas the Colombian outcome is an annual flow measure of dropout among previously enrolled students in rural public schools, identified from a within-country calendar reform. That two distinct designs, drawing on different populations, outcomes, and sources of variation, produce effects with the same sign and a similar order of magnitude makes the convergence informative. It suggests that conflicting timing institutions, captured by school-harvest misalignment, are a quantitatively important margin for human capital accumulation. More broadly, the evidence shows that institutions shape economic outcomes not only through their content, but also through their timing.

This study contributes to two broad strands of literature. First, we contribute to the literature on the economic effects and persistence of institutions. A large literature studies how institutional content—including property rights, legal systems, and political authority—shapes economic outcomes (North, 1991; Acemoglu et al., 2001; Glaeser et al., 2004; Rodrik et al., 2004; Acemoglu

and Johnson, 2005) and how institutional arrangements persist over time (Banerjee and Iyer, 2005; Nunn, 2008; Dell, 2010; Nunn and Wantchekon, 2011; Michalopoulos and Papaioannou, 2013, 2016; Lowes et al., 2017). Considerably less is known about the temporal structure institutions impose on economic behavior. A small but growing literature shows that fixed institutional schedules can matter when they interact with local economic conditions. Montero and Yang (2022), for example, show that patron-saint festivals in Mexico have persistent economic consequences when their fixed dates coincide with planting or harvest periods, while other work documents how school calendars affect parental labor supply, particularly among mothers (Price and Wasserman, 2024; Graves, 2013; Duchini and Van Effenterre, 2022).

We contribute to this literature by showing that institutional timing is quantitatively important for children’s human capital accumulation and that timing institutions can themselves be inherited and persist, generating misalignment between institutional schedules and local economic seasonality. We also contribute empirically by linking historical institutional origins to contemporary household behavior through complementary cross-country and within-country designs. Although prior work has examined the effects of instructional time and school schedules (Pischke, 2007; Marcotte, 2007; Goodman, 2014; Bostwick et al., 2022; De Roux and Riehl, 2022), to the best of our knowledge, our study is the first to link inherited calendar design to local economic conditions, and analyze school-harvest consequences using both cross-country and within-country variation.³

Second, we contribute to the education and child labor literature by showing that calendar misalignment is an important and relatively underexplored determinant of school dropout in agrarian settings (Basu and Van, 1998; Baland and Robinson, 2000; Edmonds and Pavcnik, 2005; Beegle et al., 2006; Edmonds, 2007). Existing work largely treats schooling decisions as driven by income shocks and liquidity constraints (Watkins, 2000). The studies closest to our focus examine how the school calendar interacts with agricultural seasonality along particular margins. Dillon (2021) shows that a shift in Malawi’s school calendar changed the timing of school-related

³De Roux and Riehl (2022) exploit the same school calendar reform to study a different question, namely how the resulting academic break between high school and college affected college enrollment and early-career earnings.

expenses relative to the harvest, forcing poor households to sell crops early at low prices, and Ito and Shonchoy (2026) show that annual exams coinciding with the harvest season raise school dropout among children from agricultural households in South Asia. Two concurrent working papers additionally study related questions. Merfeld (2023) shows that harvest-school overlap mediates the effect of rainfall shocks on child labor across a large set of developing countries. Allen (2024) uses a calendar reform in Malawi in a shift-share design and finds that greater overlap between school and farming calendars reduces grade advancement, corroborating the schooling costs of misalignment in a different setting.

Our paper contributes in three margins. First, we study the origins of misalignment. Our cross-country analysis documents that calendar-harvest overlap is systematically higher in former colonies, traces it to the inheritance of the colonizers' calendars, and establishes a causal link between overlap and dropout worldwide. The Colombian reform complements by showing that domestic standardization across heterogeneous agriculture produces the same conflict without any colonial inheritance. Second, the Colombian reform unified a dual public school calendar system into a single national calendar, creating a direct and contemporaneous counterfactual. It also generates two-sided variation, since the same policy increased overlap for some schools and reduced it for others. This allows us to show that outcomes deteriorate where alignment worsened and improve where it strengthened, a sign reversal that one-directional designs cannot deliver and that ties the effects to the overlap channel itself. Third, our estimates come from the administrative census of schools, covering the universe of rural public schools over more than ten years. Taken together, the consistency of the results across distinct populations, outcomes, and sources of identifying variation strengthens the interpretation that institutional timing is an important margin of human capital accumulation.

The remainder of the paper is organized as follows. Section 2 documents systematic misalignment between school and agricultural calendars across countries, while Section 3 links it to colonial institutional legacies, and presents instrumental variable estimates of the effect of calendar overlap on school dropout. Section 4 describes the institutional setting and introduces

the calendar reform in Colombia. Section 5 presents the data of the natural experiment. Section 6 outlines the empirical strategy. Section 7 presents the main school-level results with their corresponding robustness checks, and Section 8 presents evidence on how the calendar reform affected children’s agricultural work. Finally, Section 9 concludes.

2. Worldwide Evidence on School–Harvest Calendar Overlap

2.1. School Calendars Across Latitudes

School calendars were institutionalized during the 19th century, when European states formalized public primary education through foundational school laws that established centralized, state-regulated systems. These laws created the administrative apparatus through which the structure of the school year, including its term dates and vacations, was standardized and enforced (Gold, 2002; Lelièvre, 2021).⁴

Attendance concerns pushed school calendars toward minimizing conflicts with agricultural cycles, with instruction concentrated in periods of low agricultural activity (Pedersen, 2012). As a result, the academic year typically began after the summer, allowing students and their families to participate in harvest activities.⁵ Since then, and until current times, most Northern Hemisphere countries have adopted a school calendar that begins in September or October, following the summer, and ends around May or June.

However, the notion of “summer” does not translate uniformly across latitudes. Equatorial countries lack clearly defined seasons, producing substantial within-country variation in climate and harvest schedules. Countries in the Southern Hemisphere experience summer from December to February, so their agricultural seasons run out of phase with those in the north. School calendars

⁴Particularly, France enacted the Loi Guizot in 1833, Belgium the Loi Nothomb in 1842, Spain the Ley Someruelos in 1838, the Netherlands the Schoolwet in 1806, and the United Kingdom the Forster Act in 1870. Compulsory attendance legislation came later, including the Mundella Act (1880) in the UK and the Jules Ferry Laws (1882) in France, but by then the institutional template of the school calendar had already been established

⁵Additional evidence suggests that the emergence of summer breaks was also influenced by the aristocratic hunting season in France (Lelièvre, 2021), as well as religious festivities, urban social habits, budgetary constraints, and the preferences of school officials in the United States (Gold, 2002; Weiss and Brown, 2003).

should therefore vary depending on the latitude of the country.

In many equatorial and southern countries, nonetheless, school calendars do not follow local geography. Colonial inheritance often overrides it, altering school calendars while leaving harvesting schedules unchanged. To document this we assemble a worldwide dataset of current school and harvesting calendars.⁶ Appendix Figure A.1 plots the share of monthly school days (i.e., the school calendar) by country sorted by latitude and colonial origin.⁷ Many countries near the Equator and in southern latitudes follow the northern calendar despite their latitude. The southernmost ones, below the Tropic of Capricorn, do move their long vacation to December and January, their own summer. Most of the rest were historically colonized and inherited their colonizer's calendar rather than one suited to local conditions. Harvest timing, in contrast, tracks geography closely.⁸

Table 1 tests these patterns formally. In column (1) we regress an indicator for adopting a Northern Hemisphere summer vacation on latitude, defined as positive in the north and negative in the south. Moving further north significantly increases the likelihood of scheduling school vacation during July and August, so geography does shape calendars.⁹ In column (2) we regress the same outcome on a dummy for the country being formerly a colony. Formerly colonized countries are on average less likely to adopt the northern summer schedule.

This unconditional difference, however, reflects location, since most former colonies lie near the Equator or in the southern hemisphere, where a northern summer break is climatically unsuitable. In Column (3) we run a regression conditioned on latitude, and this relationship reverses. A former colony is 19 percentage points more likely to schedule its main vacation

⁶School calendars were hand-collected from multiple available sources. Harvesting calendars, proxied by the country's crop with the largest harvested area, were constructed based on data from Sacks et al. (2010). Additional details on the construction of both calendars are provided in Appendix B.1.

⁷Lai et al. (2022) compile a global dataset of school and public holidays for more than 200 countries. Our dataset records instructional and vacation days at the daily level and is built to be matched against harvest calendars, yielding a measure of school-harvest alignment rather than of holiday timing.

⁸Appendix B.2 documents and discusses both patterns. Appendix Table B.1 summarizes the contrast in ten-degree latitude bands. Harvest timing follows latitude (Appendix Figure B.1a). School vacations do not (Appendix Figure B.1b). Never-colonized countries place their vacation near the harvest, while in former colonies the two drift apart, most of all south of the equator.

⁹We exclude Liberia, which was settled by freed African Americans and never colonized by a European power. It adopted a US-style school calendar, so it fits neither our colony nor our never-colonized category.

during the northern summer than a never-colonized country in the same location. Senegal and Thailand illustrate this pattern. Both lie at nearly the same latitude. Senegal, a former French colony, schedules its long vacation during the northern summer, mirroring the French calendar. Thailand, never colonized, places its long break during its own hottest season. This conditional difference is consistent with inheriting the colonizers' calendars.

Table 1: Adoption of the Northern-Hemisphere Summer Vacation

Dependent variable: Main vacation in NH summer			
	(1)	(2)	(3)
Latitude	0.012*** (0.001)		0.014*** (0.001)
Colony		-0.317*** (0.074)	0.190** (0.087)
Observations	120	120	120
Mean dep. var.	0.683	0.683	0.683
R-squared	0.380	0.102	0.400

Notes. The table reports OLS regression estimates across 120 countries. The dependent variable is a binary indicator equal to 1 if a country's primary school vacation falls during the Northern Hemisphere summer (July–August), constructed from national Ministry of Education administrative calendars. Latitude is measured in signed degrees of latitude, positive north of the Equator and negative for south. Colony is an indicator variable equal to 1 for countries historically colonized by Belgium, France, the Netherlands, Portugal, Spain, or the United Kingdom (colonizing powers are coded as 0). Robust standard errors are reported in parentheses. ***, **, * denote statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

2.2. Colonial Origins of the School and Harvesting Calendar Overlap

To further understand the role of colonialism on influencing school calendars, Figure 1 plots the monthly share of school days in former colonies and compares them with those of their colonizers (plotted in the first row of the figures). Some colonial calendars persist despite substantial differences in climate and agricultural cycles. Countries colonized by France, the United Kingdom, Belgium, and the Netherlands show a high degree of persistence relative to the school calendar of their colonizer. In contrast, Spanish and Portuguese colonies appear to have adapted their school calendars to southern latitudes. These patterns suggest that school calendars

in some southern countries were shaped more by the institutional practices of their colonizers than by local agricultural conditions.

The extent to which colonial calendars persist seems to depend on the degree of institutional autonomy that former colonies had when their school calendars were established. For example, Spanish and Portuguese colonies, which generally gained independence earlier, subsequently developed calendars that were less similar to those of their colonizers. In contrast, countries that remained under French, British, or Dutch rule after centralized education systems had been established exhibit greater calendar similarity today. These patterns suggest that the timing of independence relative to educational institutionalization is systematically associated with the persistence of inherited calendar structures.

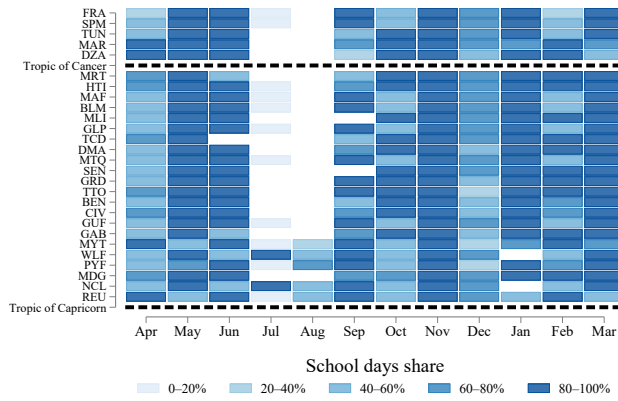
Figure 2 plots school calendar similarity between former colonies and their colonizers against the years between the colonizer's education law and the colony's independence.¹⁰ Calendar similarity to the colonizer rises with years a colony spent under colonial rule after the colonizer built its state education system, a proxy for exposure to the colonizer's educational institutions.

Two notable examples of this mechanism are Spain and France. Most Spanish colonies, including Colombia, gained independence well before educational systems were formally institutionalized in Spain (in 1838). This early independence may have enabled these countries to establish their own calendars rather than inherit those of the colonizer. In contrast, former French colonies gained independence only after school systems had already been institutionalized in France and were therefore more likely to retain calendar structures that closely mirror those of the former colonizer. These examples highlight how the timing of independence shapes the alignment between school calendars and local conditions, driving the broader pattern shown in Figure 2.

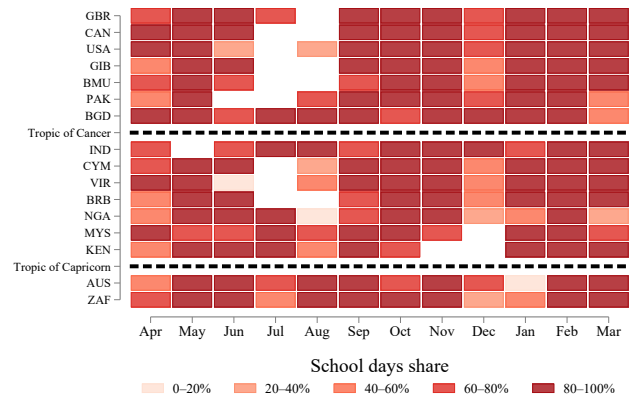
¹⁰ Similarity between both school calendars is computed as the share of the year during which students in both the colony and the colonizing country are simultaneously on similar school vacation.

Figure 1: Monthly School-Day Shares by Colony and Colonizer

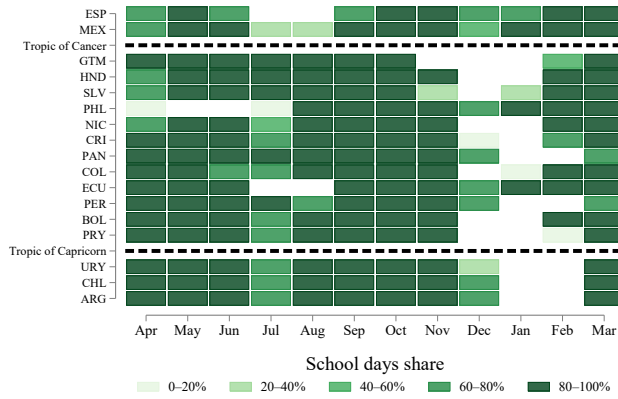
(a) France



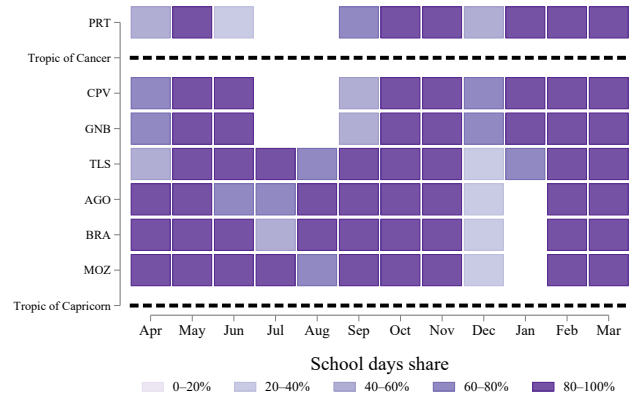
(b) United Kingdom



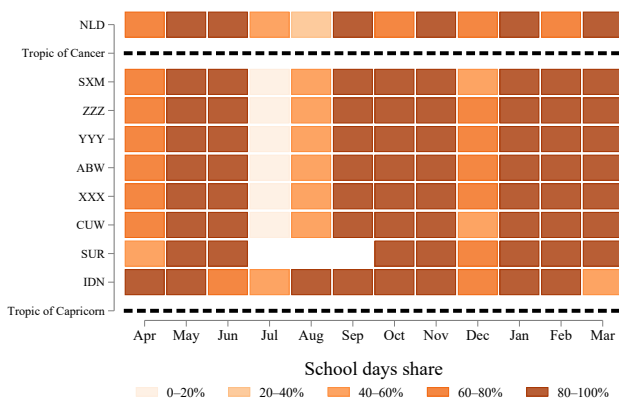
(c) Spain



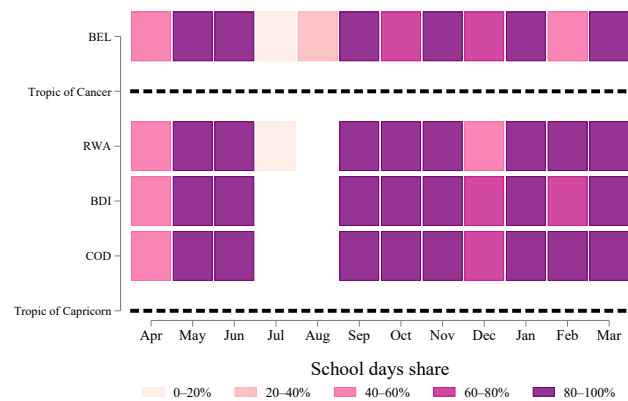
(d) Portugal



(e) The Netherlands

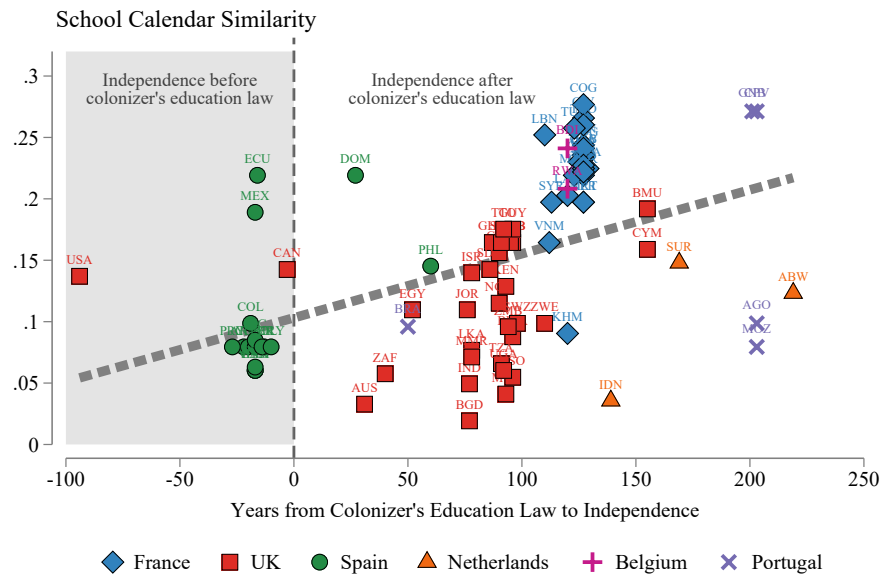


(f) Belgium



Notes: The figure plots the monthly share of instructional days across the 12-month calendar (April to March) for former colonies, organized by colonizing power: (a) France, (b) United Kingdom, (c) Spain, (d) Portugal, (e) The Netherlands, and (f) Belgium. In each panel, the former colonizing power is listed at the top, and former colonies are arranged vertically by latitude. Dashed horizontal lines mark key geographic boundaries (Tropic of Cancer and Tropic of Capricorn). Shading indicates the proportion of school days in a given month. For each country, the information corresponds to the most recent school calendar data available. Source: Own elaboration based on official sources.

Figure 2: Colonial Legacy and Similarity Between School Calendars of Colonies and Colonizers



Notes: The y-axis plots school calendar similarity, defined as the proportion of days in a calendar year during which both the former colony and its historic colonizer are simultaneously on the same school vacation. The x-axis measures institutional exposure, calculated as the difference in years between a colony's independence date and the enactment year of the colonizer's foundational education law. Positive values indicate that the colony gained independence after the colonizer institutionalized its education system, so larger values correspond to longer exposure to the colonizer's educational framework. Countries with missing independence or education-law dates are excluded. For territories that have not become independent, we use the present year, so their values reflect uninterrupted exposure through today. Solid line indicates a linear fit with a 95 percent confidence interval. Source: Own elaboration based on official and historical sources.

3. School–Harvest Overlap and Dropout: Colonial Inheritance

3.1. Inherited Calendars and School–Harvest Overlap

To measure the conflict of inherited school calendars with local agricultural cycles, we compute the school–harvest calendar overlap $O_i \in [0, 1]$ as the share of days in the year on which the school and harvest calendars overlap in each country i :

$$O_i = \frac{|S_i \cap H_i|}{|\mathbb{D}|},$$

where $\mathbb{D}$ denotes the set of days in a calendar year, $S_i \subseteq \mathbb{D}$ denotes the set of days when school is in session in country i , and $H_i \subseteq \mathbb{D}$ denotes the set of days corresponding to the harvesting season of country i 's main crop. The cardinality in the intersection of these two sets, $|S_i \cap H_i|$, captures the number of days in which students are required to attend school during periods of high agricultural activity. On average, both calendars overlap 15 percent of the year, corresponding to roughly 54 days.¹¹

School–harvest overlap is systematically higher in former colonies independent of their location. Table 2 regresses O_i on an indicator for colonial status and on latitude. Overlap is significantly higher in former colonies conditional on latitude, and significantly lower at northern latitudes. Both estimates are robust to alternative sets of controls.

This excess overlap among former colonies is related to inherited school calendars. To evaluate this, we measure the misalignment between the local harvest and the inherited colonial calendar as the number of months between a colony's main crop harvest and the colonizer's vacation period. Figure 3 plots the school–harvest overlap, O_i , against this distance. Among former colonies whose calendars remain closest to their colonizer's, O_i rises sharply with the distance ($\beta = 0.053$, $p < 0.05$). Among colonies with less calendar similarity, the relationship is flat

¹¹Appendix Table A.1 provides descriptive statistics of this measure.

Table 2: School–Harvest Overlap by Colonial Status and Latitude

	Dependent variable: School-Harvest Overlap					
	(1)	(2)	(3)	(4)	(5)	(6)
Colony	0.091*** (0.016)		0.061*** (0.023)	0.065*** (0.024)	0.066*** (0.025)	0.059** (0.025)
Latitude		-0.002*** (0.000)	-0.001* (0.000)	-0.001** (0.000)	-0.001* (0.000)	-0.001* (0.000)
Observations	120	120	120	120	120	120
Mean dep. var.	0.15	0.15	0.15	0.15	0.15	0.15
Macroeconomic controls	No	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	No	Yes	Yes
Trade controls	No	No	No	No	No	Yes

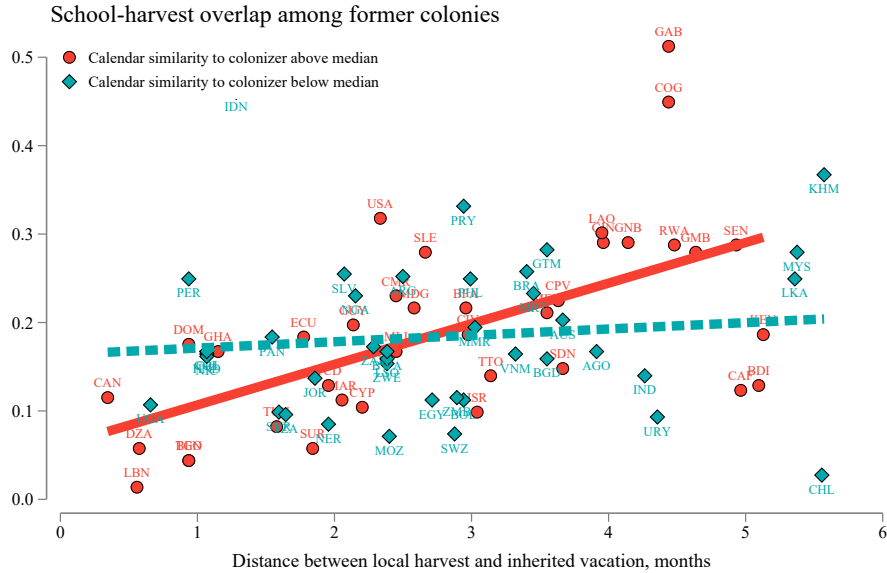
Notes. Columns 1 and 2 include the colony indicator and latitude separately. Columns 3 to 6 include both and add the control sets sequentially, in the same order as in Table 3, where latitude enters as the geographic control. Macroeconomic controls: GDP, consumption share of GDP, and agricultural value added. Demographic controls: population. Trade controls: exports and the total trade volume between each country and the set of colonizers, comprising Belgium, France, Portugal, Spain, the Netherlands, and the UK. Robust standard errors in parentheses. ***, **, * denote significance at the 1, 5, 10 percent significance level. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and [Sacks et al. \(2010\)](#).

($\beta = 0.012$, statistically insignificant).¹² Students in a colony with an inherited calendar whose harvest falls five months from the inherited vacation spend about three more months in school during the harvest than students in a colony whose harvest coincides with it. Among colonized countries, therefore, retaining the colonizer’s calendar turns a distant harvest into school–harvest conflict.

This increased school–harvest overlap can have important consequences for schooling, particularly among resource-constrained agricultural households. During peak agricultural periods, households face a sharp increase in labor needs for activities such as harvesting, crop collection, sorting, processing, storage, transportation, and other related farm tasks. Economically marginalized households, who are often credit-constrained, are unable to hire sufficient outside labor and therefore rely more heavily on family labor, including children. When these seasonal labor demands coincide with the school term, children face competing demands on their time, increasing the opportunity cost of school attendance and potentially disrupting continued enroll-

¹²The difference between the two slopes is significant at the 5 percent level. Appendix Table A.2 reports the regression estimates with colonizer fixed effects in column (4), with additional estimates in other columns.

Figure 3: Inherited Vacation, Local Harvest, and School-Harvest Overlap



Notes: Each point is a former colony. The horizontal axis is the circular distance in months between the midpoint of the colony's main crop harvest and the midpoint of the longest vacation in its colonizer's school calendar, computed treating the year as a circle. Colonies are split by the share of the colonizer's vacation days on which the colony is also on similar vacation, above or below the median among former colonies. Lines are linear fits within each group. Territories with no agricultural production are excluded because their harvest midpoint is undefined. Source: Own elaboration based on data described in Appendix B.1.

ment. To examine this pattern, Figure 4a correlates school–harvest overlap to the country-level out-of-school share, a close proxy of school dropout.¹³ Consistent with this mechanism, countries with greater overlap between school and agricultural calendars have a larger share of children out of school.

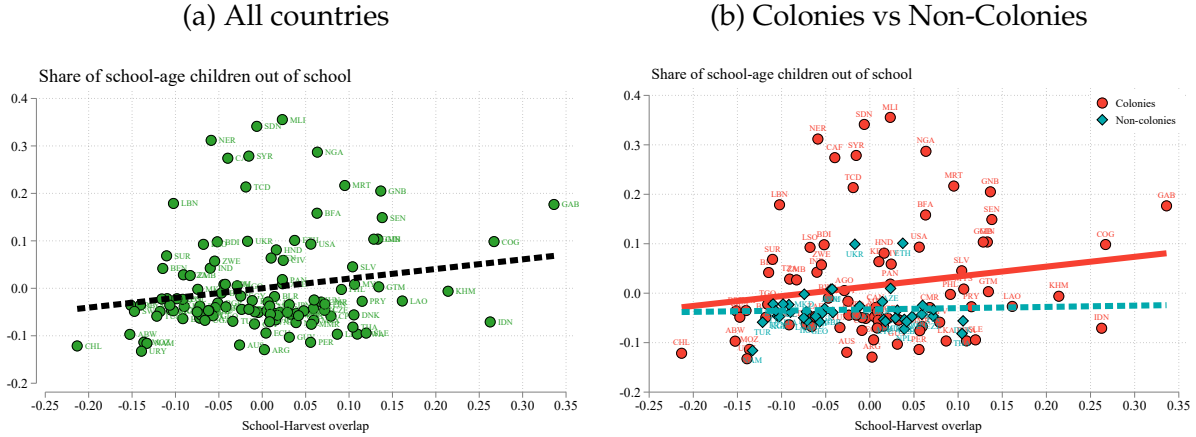
This positive relationship, nonetheless, is almost entirely driven by countries with a colonial legacy. Figure 4b separates countries with and without a colonial past, revealing a steeper gradient among former colonies and little to no correlation among countries with no colonial history.¹⁴

These correlations should not be interpreted as causal. Unobserved factors may influence both calendar structure and educational outcomes, generating omitted variable bias. In particular, low institutional quality, itself a legacy of colonialism, may drive both poor calendar design and higher dropout rates, so the colony coefficient in Table 2 and the gradients in Figure 4b likely

¹³We condition on latitude and GDP to isolate variation that is not driven by geography and income levels.

¹⁴Appendix Table A.3 shows this relationship when varying the set of controls used.

Figure 4: School Dropout and the Overlap Between School and Harvesting Calendars



Notes: Each panel plots the share of primary-school-age children out of school against the overlap between school and harvesting calendars, with both variables residualized on latitude and GDP. Residuals are computed from pooled regressions on the common estimation sample. In Panel **b**, fitted lines are estimated separately for former colonies and countries without a colonial past. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and [Sacks et al. \(2010\)](#).

absorb variation in a wide range of colonial institutional features beyond the school calendar ([Rodrik et al., 2004](#); [Acemoglu and Johnson, 2005](#)). The differential gradient is consistent with our interpretation but does not uniquely identify it, as countries with high overlap may simply have lower institutional capacity in general, or non-colonized countries with high overlap may have compensating advantages that attenuate the effect. In addition, dropout rates could themselves affect the design of school calendars, creating the potential for reverse causality.

3.2. Causal Estimates: IV regression

To estimate causal effects, we exploit historical colonial legacies and instrument school–harvest overlap in former colonies using the coincidence between the school calendar of the colonizing country and the agricultural calendar predicted by latitude. Let $c(i)$ denote the country that colonized country i and let $\mathbb{L}_i$ denote the set of countries located in the same latitude zone as country i .¹⁵ We construct the following instrument:

¹⁵Latitude zones are defined as: Equatorial (0–10°), Tropical (10–23.5°), Subtropical (23.5–35°), Temperate (35–50°), and Cool Temperate (50–66.5°).

$$Z_i = \frac{1}{|\mathbb{D}|} \sum_{d \in \mathbb{D}} S_{c(i),d} \hat{p}_{id},$$

where $S_{c(i),d}$ is an indicator equal to one if schools are in session in the colonizing country $c(i)$ on day d , $\hat{p}_{id} \in [0, 1]$ is the predicted probability that a country with the agricultural conditions of country i is harvesting on day d , and $\mathbb{D}$ denotes the days of the year. We compute $\hat{p}_{id}$ as the share of other countries in the same latitude zone $\mathbb{L}_i$ and with the same main crop as i that harvest on day d . The instrument is thus the expected share of the year during which the colonizer's school calendar coincides with predicted harvesting activity.¹⁶ For countries that were not colonized the instrument takes the value of zero. On average, both calendars overlap 12 percent of days in the year (see Appendix Table A.1).

Using this instrument, we estimate the following equations by two-stage least squares:

$$\begin{aligned} O_i &= \alpha_0 + \alpha_1 Z_i + \mathbf{X}_i' \boldsymbol{\pi} + u_i \\ Y_i &= \beta_0 + \beta \hat{O}_i + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i, \end{aligned} \tag{1}$$

where $Y_i \in [0, 1]$ is the share of primary-school-age children who are not enrolled in primary or secondary education in country i , our proxy for school dropout.¹⁷ $\hat{O}_i$ denotes the predicted overlap between school and agricultural calendars, and $\mathbf{X}_i$ denotes a vector of country level controls, included in both stages.¹⁸ Standard errors are heteroskedasticity-robust.

¹⁶ $\hat{p}_{id}$ is constructed as a leave-one-out estimate of the probability that a country harvests on a given day of the year. For each day d , we estimate the probability that a country growing the same main crop as country i and located in the same latitude zone is actively harvesting on day d , using all countries in that peer group except country i itself. Formally, for each cell combination of country j and day d we compute $\hat{p}_{id} = \sum_{j \in \mathbb{L}_i^c \setminus \{i\}} \mathbf{1}[\text{harvest}_{jd}] / (|\mathbb{L}_i^c| - 1)$, where $\mathbb{L}_i^c \subseteq \mathbb{L}_i$ denotes the set of countries sharing both the same main crop and the same latitude zone as country i . When the peer group contains fewer than two countries, we progressively widen the comparison group first to all countries in the same hemisphere growing the same crop, and then to all countries growing the same crop globally. The resulting daily measure $\hat{p}_{id} \in [0, 1]$ reflects the predicted probability of harvest on each day of the year for a country with similar agroclimatic conditions to country i , purged of country i 's own harvest schedule to avoid mechanical correlation.

¹⁷ Primary-school age is defined by each country's official entrance age and duration of primary education so that the corresponding age range varies across countries.

¹⁸ Controls include (i) latitude, (ii) GDP, the consumption share of GDP, and agricultural value added, (iii) population, and (iv) exports in USD and the total trade volume between each country and the set of colonizers, comprising Belgium, France, Portugal, Spain, the Netherlands, and the UK.

The coefficient $\hat{\beta}$ in Equation 1 is consistently estimated if the instrument Z_i satisfies the exclusion restriction. This requires that the school calendars of former colonial powers affect school dropout in their former colonies only through the school calendars that these colonies inherited, and not through any other channel. This assumption is plausible because colonial powers originally designed their domestic school calendars to reflect their own climatic and agricultural conditions rather than those of the territories they later administered, making the timing of school terms in the colonizing country unlikely to be directly related to contemporary educational outcomes in former colonies located at very different latitudes.

The instrument further combines this colonial school calendar with harvest timing predicted by latitude peers, which captures the agricultural conditions that country i would likely face in the absence of colonial institutional influence and prevents the instrument from being mechanically correlated with country-specific agricultural patterns. Together, these two components isolate variation that is specific to calendar timing and should be orthogonal to other dimensions of colonial institutional inheritance such as property rights, legal origin, or state capacity (Rodrik et al., 2004; Acemoglu and Johnson, 2005).

Table 3 presents the results of estimating Equation 1.¹⁹ The estimated coefficients are positive and statistically significant across specifications. Columns 1 to 4 of Panel A imply that a one percentage point increase in the overlap between school and agricultural calendars raises the out-of-school share by approximately 0.77 to 0.79 percentage points. The instrument is relevant, with effective first-stage F-statistics between 11.2 and 19.6 across specifications (Montiel Olea and Pflueger, 2013). We also report confidence sets based on the Anderson and Rubin (1949) test, which remain valid regardless of instrument strength. These sets exclude zero in every specification of both panels, with lower bounds between 0.05 and 0.38.

One margin of the instrument nonetheless deserves separate scrutiny. Because Z_i equals zero for countries that were never colonized, the full-sample estimates draw in part on the comparison between former colonies and never-colonized countries, a contrast that may correlate with broader

¹⁹The first stage results are presented in Appendix Figure A.2.

Table 3: Effect of School–Harvest Overlap on the Out-of-School Share

	Dependent variable: Dropout rate				
	(1)	(2)	(3)	(4)	(5)
Panel A: Full sample					
School-Harvest overlap	0.786*** (0.280)	0.781*** (0.250)	0.772*** (0.243)	0.785*** (0.251)	0.558** (0.282)
Effective first-stage F	19.58	18.47	18.74	17.49	12.56
Partial first-stage R^2	0.230	0.227	0.241	0.238	0.226
AR 95% confidence set	[0.36, 1.66]	[0.38, 1.54]	[0.38, 1.50]	[0.38, 1.55]	[0.13, 1.56]
Observations	120	120	120	120	120
Panel B: Former colonies					
School-Harvest overlap	0.484* (0.289)	0.591** (0.293)	0.590** (0.285)	0.630** (0.305)	
Effective first-stage F	12.81	12.19	12.61	11.18	
Partial first-stage R^2	0.260	0.248	0.272	0.256	
AR 95% confidence set	[0.05, 1.53]	[0.15, 1.66]	[0.16, 1.61]	[0.18, 1.79]	
Observations	81	81	81	81	
Geographic controls	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	No	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes	Yes
Trade controls	No	No	No	Yes	Yes
Colonial status control	No	No	No	No	Yes

Notes. The dependent variable is the out-of-school share, the share of primary-school-age children not enrolled in primary or secondary education. Primary-school age follows each country's official entrance age and duration of primary education, so the age range varies across countries. For each country, we use the latest available value during the 2010–2019 period. Geographic controls: country latitude. Macroeconomic controls: GDP, consumption share of GDP, and agricultural value added. Demographic controls: population. Trade controls: exports and the total trade volume between each country and the set of colonizers, comprising Belgium, France, Portugal, Spain, the Netherlands, and the UK. The colonial status control is an indicator for former colonial status, included in both stages. It applies to Panel A only, since it is constant within the sample of former colonies and therefore drops in Panel B, which has four columns. The effective first-stage F is the statistic of [Montiel Olea and Pflueger \(2013\)](#), whose critical value for a worst-case bias of 10 percent is 23.1 with a single instrument. The partial first-stage R^2 is the share of the variation in school–harvest overlap explained by the instrument after partialling out the controls. The AR 95% confidence set contains the values of the coefficient not rejected at the 5 percent level by a heteroskedasticity-robust [Anderson and Rubin \(1949\)](#) test, and is valid regardless of instrument strength. ***, **, * denote significance at the 1, 5, 10 percent significance level. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and [Sacks et al. \(2010\)](#).

institutional legacies. We address this in two ways. First, column (5) of Panel A adds an indicator for colonial status to both stages, so that the estimate no longer draws on the difference between colonized and never-colonized countries. The coefficient falls to 0.56, remaining positive and precisely estimated. Second, Panel B restricts the sample to former colonies, where identification relies entirely on variation across the degree of calendar mismatch rather than on the comparison between colonized and non-colonized countries, and the coefficients range from 0.48 to 0.63. We treat these estimates as a more conservative benchmark, since they are immune to any confound operating through colonial status itself.

All estimates in both panels remain stable across different sets of controls. The estimates

are also robust to colonizer, region, and latitude-zone fixed effects and controlling for religious fractionalization, with Anderson–Rubin confidence sets that exclude zero in every specification (Appendix Table B.2). For additional robustness, Appendix B.3 considers a second instrument that combines the interaction between the school calendar of the colonizing country, the agricultural calendar predicted by their latitude, and the degree of exposition to colonial education institutions. This second instrument resembles a shift-share design with both the *shift* and the *share* components being arguably orthogonal, yielding nearly identical estimates to our primary specification.

4. A Natural Experiment: Colombia’s School Calendar Reform

The cross-country evidence shows that overlap between school and harvest calendars matters and may be historically inherited. However, for three reasons, it cannot identify precisely who is affected or through which mechanisms. First, the instrument exploits variation that originates in colonial institutional inheritance, so the estimates speak to countries whose school calendars were shaped by colonial institutions. Second, they rely on national averages, which mask substantial heterogeneity in agricultural cycles, school calendars, and schooling decisions. Third, both calendars are measured with error at the country level, since harvesting seasons and school schedules vary across regions within a country. Within-country variation addresses the last two directly, by measuring both calendars at the level at which schooling decisions are made. It also speaks to the first by studying misalignment created by a domestic reform rather than inherited from a colonizer, which tests whether the effect extends beyond colonially shaped calendars.

We therefore complement the cross-country analysis with micro-level evidence from a natural experiment in Colombia, where a school calendar reform allows us to trace the effect of calendar misalignment while holding other institutional features constant. Although Colombia was colonized by Spain, it gained independence in 1819, well before Spain institutionalized its national

education system.²⁰ Colombian authorities therefore designed their own school calendars, which makes the Colombian case distinct from the colonial-inheritance channel emphasized in the cross-country analysis.

After independence, Colombian school calendars varied across regions, and in 1951 this variation was formally recognized as two parallel national calendars. Public schools in Nariño and Valle del Cauca followed the Northern Hemisphere pattern, while the rest of the country followed the Southern Hemisphere pattern. Colombia lies near the Equator, so neither pattern perfectly fits its agricultural cycles. A school calendar reform in 2010 unified the public school calendar across all departments. The change therefore increased school–harvest overlap for some rural public schools, reduced it for others, and left it essentially unchanged for the rest. This two-sided variation is rare and lets us test the mechanism directly, since outcomes should worsen where overlap rose and improve where it fell. The reform also illustrates a second route to misalignment, as standardizing a domestic school calendar across heterogeneous harvest schedules produces the same school–harvest conflict as inheriting a calendar built for a foreign country. We detail the reform and the resulting school–harvest conflict in what follows.

4.1. The Colombian Education System and the Dual School Calendar

Colombia’s education system consists of primary education (grades 1–5), lower secondary education (grades 6–9), and upper secondary education (grades 10–11). Schooling is provided by both public and private institutions. In 2010, approximately 78 percent of students were enrolled in public schools, and about 24 percent attended schools located in rural areas (DANE, 2011).

Historically, Colombian public schools operated under two academic calendars. *Calendar A* ran from January to November, similar to calendars in the Southern Hemisphere. *Calendar B* ran

²⁰We date the institutionalization of Spain’s education system to the *Ley de Instrucción Primaria* of 21 July 1838 (*Ley Someruelos*), which established the first functioning liberal primary school system in Spain, requiring municipalities to maintain public schools and introducing teacher certification and provincial supervision. This framework was later consolidated by the *Ley de Instrucción Pública* of 9 September 1857 (*Ley Moyano*), which introduced compulsory and free elementary schooling. Since most Spanish colonies, including Colombia, gained independence in the 1810s and 1820s, they did so before either law was enacted, leaving post-independence authorities greater scope to design their own school calendars.

from September to June, similar to calendars in the Northern Hemisphere. Both require 40 weeks of instruction per academic year ([Congreso de la República de Colombia, 1994](#)). The dual-calendar system emerged from regional differentiation in school timing and was formally recognized in 1951. Public schools in Nariño and Valle del Cauca followed the September–June calendar, while most public schools elsewhere followed the January–November calendar. Regulations at the time explicitly recognized climatic and local conditions in the southwest as reasons for maintaining a distinct schedule. Appendix C provides the detailed historical and legal evolution of the two calendars.

4.2. The school calendar reform

Following a decentralization reform in 2001, local education authorities were given the responsibility to organize school calendars subject to national requirements (see Appendix C for details). The September–June calendar used in Nariño and Valle del Cauca spanned two fiscal years and complicated administrative planning and budget execution. Beginning in 2011, public schools in these departments were therefore required to transition to the January–November national calendar. The reform did not modify the duration of instruction as schools continued to provide the required 40 weeks of class. It instead replaced a regionally differentiated public-school calendar with a common national schedule. Private schools retained discretion over which calendar to follow.²¹

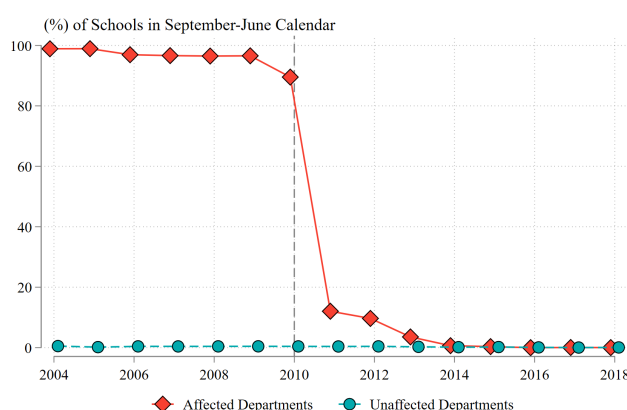
Figure 5a plots the share of public schools adopting the September–June calendar in the departments of Nariño and Valle del Cauca (hereafter, the “affected” departments) compared with the rest of the country. Prior to 2010, the period preceding the calendar reform, virtually all public schools in the affected departments operated under the September–June calendar, whereas public schools in the rest of the country consistently followed the January–November one. Beginning

²¹De Roux and Riehl (2022) provide a detailed discussion of the reform. The transition was implemented through resolutions issued by certified territorial entities under Law 715 of 2001. One example is Resolution 4143.2.21.4739 of 2008 issued by the Secretaría de Educación Municipal of Santiago de Cali ([Santiago de Cali. Secretaría de Educación Municipal, 2008](#)).

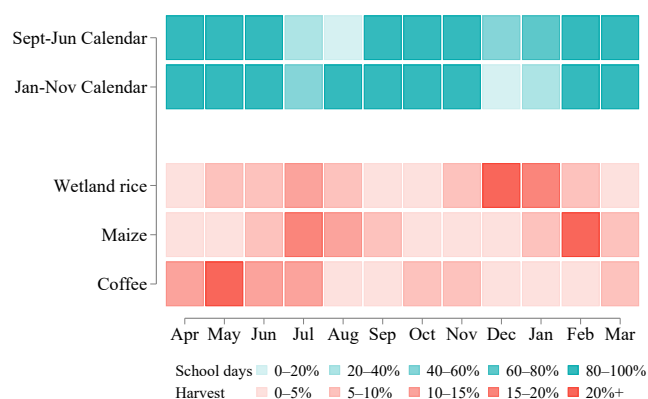
in 2011, and following the implementation of the calendar reform, public schools in the affected departments transitioned to the January–November calendar, aligning their academic schedules with those used elsewhere in the country. The transition was rapid but not instantaneous. Close to ninety percent of affected public schools switched in 2011, a share that exceeds 96 percent among the rural public schools that constitute our universe of study. The remainder followed over the next years, with full convergence by 2014.²²

Figure 5: The School Calendar Reform in Affected Departments

(a) Share of Schools under Sept.-June Calendar



(b) School Days and Harvest Intensity by Month



Notes: Panel (a) shows, for each year, the share of public schools operating under the September–June calendar. The shares are reported separately for schools in Nariño and Valle del Cauca (red dots) and for schools in the rest of the country (blue dots). The rural public schools that constitute our estimation universe transitioned faster than the figure suggests, with 96.7 percent on the January–November calendar in 2011. Panel (b) plots, in the top two rows, the proportion of school days (days of instruction) in each month as a share of total monthly workdays under each calendar, grouped in bins of 0–20, 20–40, 40–60, 60–80, and 80–100 percent. The bottom three rows plot the share of each crop’s total annual harvesting activity that occurs in each month, grouped in bins of 0–5, 5–10, 10–15, 15–20, and above 20 percent, for the three crops with the largest harvested area. Coffee refers to the stimulants category. White cells indicate zero. The plotted values correspond to the average across the affected departments of Nariño and Valle del Cauca. Source: Own elaboration based on data from the Colombian School Census (Form C-600), the Secretaries of Education of Nariño and Valle del Cauca, and the Colombian Ministry of Agriculture.

The school calendar reform induced changes in the distribution of instructional days across months. Figure 5b presents the average monthly share of instructional days under the September–June calendar prior to the reform and under the January–November calendar adopted thereafter in affected departments. The reform substantially altered the timing of school holidays throughout

²² Schools with delayed or partial transitions are excluded from our analysis. These are rare, corresponding to 4 percent of rural public schools. Our main results are robust to excluding the transition years 2011 to 2013, indicating that they are not driven by disruption during the transition itself.

the year. Following the reform, vacation time became concentrated in December and January, with a markedly shorter mid-year break in July. Whether this shift raised or lowered the overlap between school and harvest depend on the local agricultural activity of affected schools.

4.3. Agricultural Activity

Colombia is a middle-income country where around 23 percent of the population lives in rural areas and 17 percent of the workforce is employed in the agricultural sector. Agricultural production varies depending on the area of the country, with coffee, rice, and maize being the most common agricultural products. Affected departments are intensive in the production of stimulants (mainly coffee, although the category also includes tea and cocoa), rice, maize, sugarcane, and fruits and nuts, whereas the rest of the country specializes also in stimulants, maize, rice, and sugarcane but also in yams and roots. Appendix Figure A.3 plots the main crop at the *vereda* level for all the country and highlights the two affected departments. *Veredas* are the smallest administrative division in rural Colombia, with a median area of about 9 square kilometers.²³

Colombia is located very close to the Equator, implying that temperature fluctuations do not follow a seasonal pattern. Instead, temperature varies primarily with altitude. Lower-lying areas are warmer, while higher-altitude regions are colder. Despite the relative stability of temperatures throughout the year, climatic conditions do vary seasonally in terms of precipitation rather than temperature. As a result, climate seasonality is present (although to a much lesser extent than in countries with pronounced seasons) while temperature remains relatively constant over the year.

This climatic context, characteristic of equatorial countries, allows the production of a wide range of agricultural products depending on the altitude of cultivated areas. Colombia therefore

²³A *vereda* is typically formed by a grouping of land parcels bounded by geographic features and major roadways. According to the 2020 official geographic layer from the Instituto Geográfico Agustín Codazzi (IGAC), Colombia has 33,428 *veredas* with a median area of 8.9 km², ranging from 1.3 km² at the 5th percentile to 87 km² at the 95th percentile. Size varies enormously across the country, from a median of 5.8 km² in Andean departments such as Cundinamarca and Nariño to a median of 442 km² in sparsely populated southern departments such as Amazonas and Vichada.

exhibits substantial heterogeneity in both the types of crops produced and the timing of their production, which translates into considerable variation in agricultural calendars across small areas. The bottom rows of Figure 5b illustrate this for the three crops with the largest harvested area in the affected departments, coffee, wetland rice, and maize, plotting the share of each crop's annual harvesting activity that occurs in each month alongside the two school calendars. Appendix Figure A.4 presents the harvesting calendars of all other crops grown, and Appendix Figures A.5 and A.6 report them separately by department.

Wetland rice harvesting is concentrated in December and January, maize in February and around the middle of the year, and coffee between April and June. This heterogeneity is central to our analysis. Harvests follow local crops rather than a common season, so the school calendar reform did not move all schools toward or away from the harvest. Comparing the harvest rows of Panel 5b with the two calendar rows illustrates the two-sided conflict. In a region that grows mostly maize, the new calendar put school in session in August, when maize is harvested, and overlap rose. In a region that grows mostly wetland rice, it took December and January out of the school year, when rice is harvested, and overlap fell. Exposure to the reform therefore varied with the crops grown around each school.

5. Data

We combine four data sources to construct a school-level, census-wide, panel containing information on school progression, geographic location, and exposure to harvesting seasonality.

First, we use the Colombian school census for 2006–2018.²⁴ The census is collected annually by the central government through surveys completed by school principals, who report detailed information on enrollment, infrastructure, and teaching staff. For each school, principals also classify students enrolled in the previous year as promoted, repeating a grade, dropped out, or transferred to another institution. We use this information to construct our two main outcomes.

²⁴These data come from Form C-600 of the Colombian Ministry of Education and are publicly available.

The dropout rate of school s in year t is the number of students enrolled in s in year $t - 1$ who are reported as having dropped out, divided by the number of students enrolled in s in year $t - 1$. The grade promotion rate is the number promoted to the next grade divided by the same denominator. Both outcomes are school-level annual flow measures, pooled across grades and defined relative to the cohort enrolled in the previous year. Students who transfer remain in the denominator but are not classified as dropouts. We complement the census with geo-referenced information on school locations.²⁵

Second, we compute harvested areas for different crops at the *vereda* level using data from the Global Agro-Ecological Zones (GAEZ) v4, produced by the Food and Agriculture Organization of the United Nations (FAO).²⁶ The GAEZ data provide information on harvested areas for 25 crop groups, constructed on grids of approximately 9 by 9 kilometers at the equator, or about 81 square kilometers per cell. Because the median *vereda* in Colombia covers about 9 square kilometers, a typical grid cell spans several *veredas*. We assign to each *vereda* the area-weighted average of the grid cells it intersects and match the resulting crop composition to the school census. The crop information surrounding each school is therefore measured at the resolution of the grid rather than of the *vereda* itself, so nearby schools share similar measured crop portfolios. For this reason, in addition to school-level clustering, we also report standard errors clustered at the municipality level, which allow for arbitrary correlation in measured exposure across nearby schools.

Third, we use two sources on the timing of calendars. Harvesting calendars at the department-crop level come from the Rural Agricultural Planning Unit (Unidad de Planificación Rural Agropecuaria, UPRA) of the Colombian Ministry of Agriculture, and report the timing and intensity of harvesting seasons for each crop in every department.²⁷ School calendars come from administrative records of the Secretaries of Education of Nariño and Valle del Cauca, covering public schools in these departments for 2005–2008 and 2011–2015. Together they allow us to measure the overlap between the two calendars before and after the reform.

²⁵ Appendix Figure A.7 shows the affected departments and the spatial distribution of schools across the country.

²⁶ The data are available at: <https://www.gaez.fao.org/>.

²⁷ Available at: <https://www.upra.gov.co/es-co/eva/eva-2019>.

Combining these data sets, we build a school-level panel that includes a total of 31,259 schools, representing all schools in the country. Out of these, 88.6 percent correspond to public, 59.6 percent to rural, and 59.4 percent to rural, public schools (see Appendix Figure A.8). Our main analysis focuses on public schools because the reform directly changed the calendar governing these institutions, while private schools retained discretion over which calendar to follow. Within public schools we focus on rural schools, where seasonal agricultural labor demand makes the conflict between school and harvest timing most relevant, and we use urban public schools as a comparison for the agricultural mechanism in Section 7. The final estimation sample therefore corresponds to the universe of rural public schools observed in every year between 2006 and 2018, yielding a balanced panel of 18,553 schools. Appendix Table A.4 describes the sample.

Overall, public school enrollment declines over time across all departments in the sample. However, the reduction is substantially larger in the affected departments. Enrollment falls by about 9 percent in the affected departments, compared to a decline of about 2 percent in the rest of the country. Over the same period, dropout rates decrease and passing rates increase across all schools. The estimates induced by the reforms should therefore be read as a relative deterioration in affected schools against this improving counterfactual trend, rather than as an absolute increase in the raw dropout rate.

6. Empirical Strategy

6.1. Overlap Between School and Harvesting Calendars

The calendar reform modified the degree of overlap between school and harvesting calendars in the affected departments. To quantify this change, we construct a school-level measure of overlap between the two calendars in four steps:

1. Using department-level school calendars, we compute the proportion of non-instructional days in month m for each affected department d and each academic calendar $c \in \{A, B\}$.

We denote this proportion by V_{mdc} .

2. Using department- and crop-specific harvesting calendars, we identify harvesting-intensive months for each crop. We define an indicator variable, $\mathbb{1}_{\{m \in \mathcal{I}_j\}}$, equal to one if month m for crop j falls within the top quartile of the annual distribution of harvesting activity (i.e., the three most intensive months), denoted by $\mathcal{I}_j$, and zero otherwise.²⁸
3. Combining these two measures, we define the overlap $O_{sc} \in [0, 1]$ between school and harvesting calendars for each school s under calendar c as:

$$O_{sc} = \sum_{j \in J} \omega_{sj} \left(\frac{1}{12} \sum_{m=1}^{12} \max \left\{ \mathbb{1}_{\{m \in \mathcal{I}_j\}} - V_{mdc}, 0 \right\} \right). \quad (2)$$

The term inside the brackets captures the share of instructional days during harvesting-intensive months (i.e., $1 - V_{mdc}$) and is equal to zero during non-intensive months. Averaging this expression over the twelve months of the year yields, for crop j , the share of the calendar year during which instruction and intensive harvesting coincide under calendar c . We then aggregate across crops using ω_{sj} , the share of crop j 's harvested area in the *vereda* where school s is located. The resulting measure, $O_{sc} \in [0, 1]$, is the crop-weighted share of the year in which schooling and intensive harvesting overlap, defined on the same scale as the cross-country overlap measure in Section 2. Larger values indicate that more of the school year falls in periods of peak agricultural activity, and thus a higher opportunity cost of schooling.

4. Finally, we compute a school-level exposure measure, ΔO_s , defined as the change in overlap induced by the calendar reform:

$$\Delta O_s = O_{sA} - O_{sB}. \quad (3)$$

²⁸The selection of the set $\mathcal{I}_j$ might vary depending on how many months we define as harvesting intensive. We provide robustness checks in Section 7.3 varying the definition of $\mathcal{I}_j$ and show that our results do not depend on this.

A positive value of ΔO_s indicates that the post-reform January–November calendar (Calendar A) *increases* the overlap between academic activities and harvesting periods relative to the pre-reform September–June calendar (Calendar B) in school s . A negative value of ΔO_s indicates that the reform *decreases* such overlap.

Intuitively, the overlap measure captures how the reform changes the extent to which schooling competes with seasonal household labor demand. Consider a student living in a rural area where maize is the dominant crop. Under the pre-reform calendar, August falls during the school vacation, allowing children to assist with harvesting and related farm activities without conflicting with school. Under the post-reform calendar, August becomes an instructional month, as Figure 5b shows. For resource-constrained agricultural households that rely heavily on family labor and may be unable to hire sufficient outside workers during the harvest, the reform therefore creates a direct conflict between children’s schooling and household labor needs.

The change in school–harvest overlap differs substantially across crops and departments. Appendix Figure A.9 presents it for the main crops in each affected department. The overlap associated with maize harvesting, for instance, increases markedly in Nariño but decreases slightly in Valle del Cauca. Exposure to the reform therefore depends jointly on the dominant crops in each *vereda* and on department-specific harvesting calendars. Averaged across crops, the reform increased overlap by 1.7 percentage points, as shown by the overall bar at the bottom of the figure. This small average conceals substantial heterogeneity. Overlap increased in 60.4 percent of affected schools, was unchanged in 9.2 percent, and declined in 30.5 percent. This two-sided variation allows us to identify the consequences of both worsening and improving calendar alignment within a common institutional environment.

6.2. Dose-Response Design

Baseline specification. The calendar reform provides a clean source of quasi-experimental variation. It generated heterogeneity in school–harvest overlap for every affected school. We

exploit this school-level variation in treatment intensity using the following dose-response event-study specification on the panel of rural public schools:

$$y_{st} = \alpha_s + \delta_t + \sum_{k \neq 2010} \tau_k(\Delta O_s \times \mathbf{1}[t = k]) + \sum_c \gamma_{ct} X_{m(s)}^c + \varepsilon_{st}, \quad (4)$$

where y_{st} denotes the outcome of school s in year t , α_s and δ_t are school and year fixed effects, and ΔO_s is the school-level change in school–harvest overlap defined in Equation 3. For schools outside the affected departments, ΔO_s is set to zero. The coefficients of interest, τ_k , trace the effect of reform-induced changes in overlap in each year relative to 2010, the last pre-reform year.²⁹

The terms $X_{m(s)}^c$ denote pre-reform characteristics of the municipality in which school s is located—per capita GDP, conflict exposure (proxied by the number of recorded terrorist attacks), coca cultivation, and municipal public expenditure—each interacted with a full set of year fixed effects. These interactions allow schools in municipalities with different baseline economic conditions, conflict exposure, and fiscal capacity to follow different trends over time. Standard errors are clustered at the school level in the baseline specification.³⁰

To summarize the average post-reform effect and facilitate comparison with the cross-country estimates in Section 3, we also estimate a static specification that replaces the year-specific interactions with a single interaction between ΔO_s and a post-reform indicator:

$$y_{st} = \alpha_s + \delta_t + \tau(\Delta O_s \times \text{Post}_t) + \sum_c \gamma_{ct} X_{m(s)}^c + \varepsilon_{st}. \quad (5)$$

The coefficient τ provides a single estimate of the average post-reform effect per unit of change in school–harvest overlap.

Our identification relies on comparing the evolution of outcomes across schools that experienced different reform-induced changes in overlap, conditional on school fixed effects, common

²⁹ ΔO_s is measured in the same units as the overlap variable in Equation 1, with $\Delta O_s \in [-1, 1]$. When dropout is the outcome, the coefficients can therefore be expressed in the same units as the cross-country IV estimates in Section 2.

³⁰ We also report municipality-level clustering throughout as a more conservative alternative, since crop composition, and therefore treatment intensity, may be correlated across schools within the same local area.

year shocks, and the differential trends captured by the baseline municipality interactions. The identifying assumption is that, absent the reform, schools with different values of ΔO_s would have followed parallel outcome trajectories. We assess this assumption using the pre-reform event-study coefficients, reporting τ_k for $k < 2011$ in all event-study figures.

Our main specification uses standard two-way fixed effects because it is transparent and easy to interpret. With a continuous treatment and potentially heterogeneous effects, however, the two-way fixed-effects estimand need not recover a convex average of school-level treatment effects (de Chaisemartin and D’Haultfoeuille, 2020; Callaway and Sant’Anna, 2021). We therefore re-estimate the main specifications using the heterogeneity-robust estimator of de Chaisemartin et al. (2024). We also examine the sensitivity of the estimates to alternative control groups by restricting comparisons first to bordering departments and then to progressively narrower distance bands around the boundary of the affected departments. The results remain stable across estimators, clustering levels, and control-group definitions.

Finally, because the reform applied to all public schools in the affected departments, one could estimate a conventional difference-in-differences specification comparing Nariño and Valle del Cauca with the rest of the country. Such a design, however, would discard the substantial within-treatment variation in overlap, rely on only two affected departments for treatment-level inference, and use the simple average of a large and heterogeneous set of untreated schools as the counterfactual. We therefore complement the dose-response analysis with synthetic difference-in-differences (SDID) estimates (Arkhangelsky et al., 2021), which address the choice of counterfactual by constructing a data-driven weighted combination of untreated schools whose pre-reform trajectory more closely matches that of the treated schools. Whereas Equations 4 and 5 use variation in the reform-induced change in overlap to isolate the school–harvest timing channel, SDID captures the reform’s total effect on the affected departments, including components common to all treated schools. We therefore present the SDID results as a complementary robustness exercise and use them to gauge the magnitude of the reform’s overall effect.

Specification by direction of the change. The baseline specification sets the dose to zero outside the affected departments. Its coefficient therefore reflects two comparisons, between more and less exposed treated schools and between treated and untreated schools. The second comparison means that a shock hitting all affected schools, such as disruption during the transition year, could be attributed to overlap.³¹

The reform allows a sharper test of the opportunity-cost mechanism that is immune to such shocks. If the effects operate through the overlap between schooling and agricultural work, outcomes should deteriorate where overlap rose and improve where it fell. A shock common to the affected departments would instead move both groups in the same direction.

We implement this test by splitting treated schools into those with a positive change in overlap ($\Delta O_s > 0$) and those with a negative change ($\Delta O_s < 0$). For each group $g \in \{+, -\}$, we estimate

$$y_{st} = \alpha_s + \delta_t + \tau^g (|\Delta O_s| \times \text{Post}_t) + \sum_c \gamma_{ct} X_{m(s)}^c + \varepsilon_{st}, \quad \text{for } g \in \{+, -\}, \quad (6)$$

on the schools in group g together with the full set of untreated schools. The coefficient τ^g measures the effect per percentage point of overlap change in each group's own direction. We use the absolute value of the dose as treatment intensity to ensure coefficients are directly comparable across groups. We also estimate the event-study analogue of Equation 6, replacing the post indicator with year-specific interactions as in Equation 4.

7. School-Level Effects of the Reform

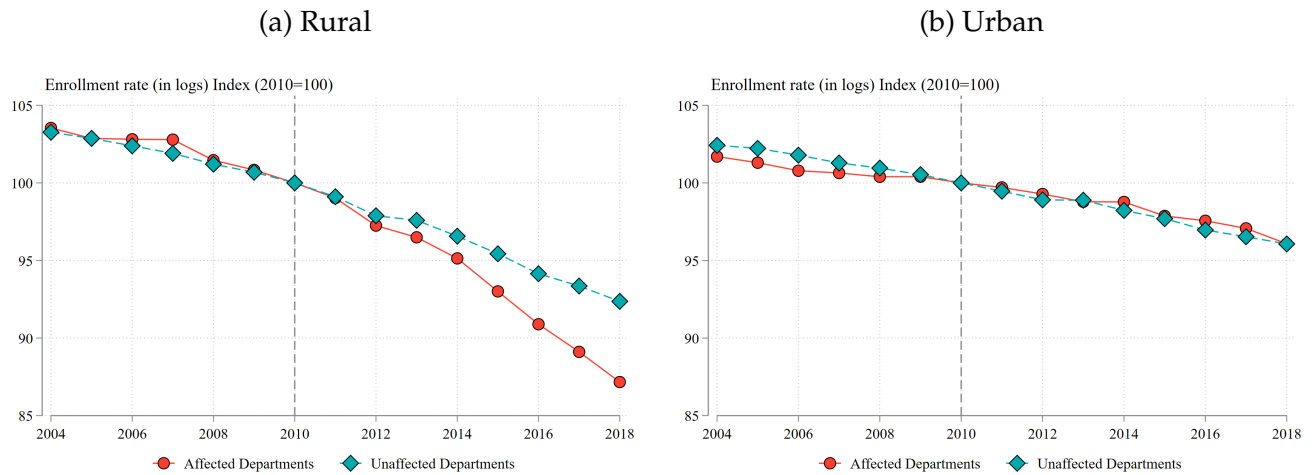
7.1. Effects of the Reform on School Progression

We begin by examining how rural public-school enrollment evolved around the reform. We plot the evolution of school enrollment, normalized to the year prior to the reform, in Figure 6.

³¹One such shock is the one-off academic break during the transition year, which De Roux and Riehl (2022) study in the context of college enrollment. We address such shocks by adding treated-by-year fixed effects, which absorb anything common to affected schools, and our estimates remain essentially unchanged.. More details are provided in Section 7.3.

We present estimates separately for rural public (Panel 6a) and urban public (Panel 6b) schools. Prior to the reform, school enrollment in affected and unaffected departments evolved in parallel, indicating no differential pre-reform trends. After the reform, we observe a sudden decline in school enrollment among rural public schools in affected departments, with the gap widening over time through 2018. We do not see any change in enrollment among urban public schools.

Figure 6: Public School Enrollment in Affected and Unaffected Departments



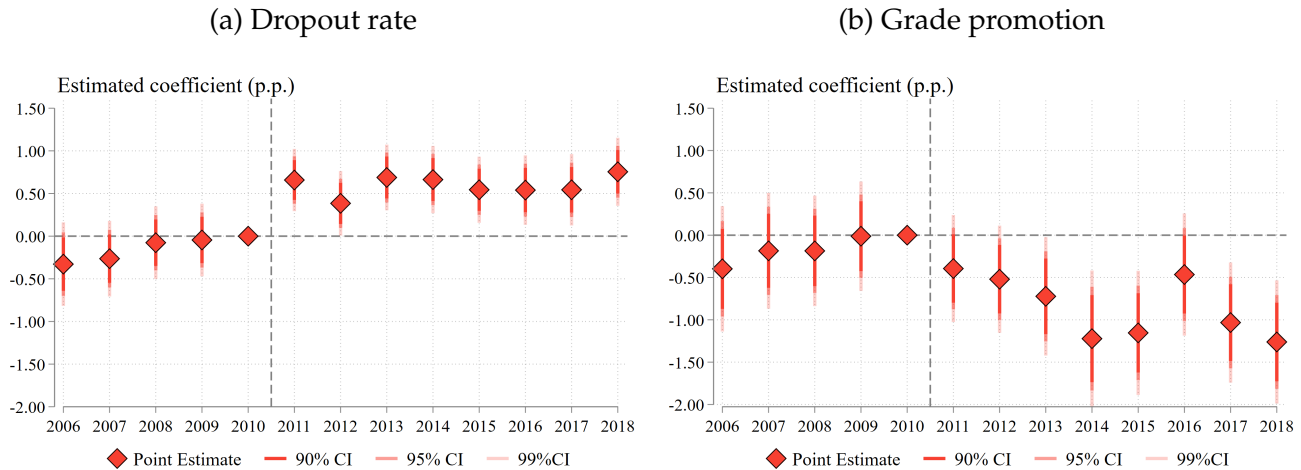
Notes: Each panel plots public school enrollment in the affected departments (Nariño and Valle del Cauca) and in the rest of Colombia, indexed to 100 in 2010, the last pre-reform year, for rural public schools (Panel a) and urban public schools (Panel b). The vertical line marks the reform. Source: Own elaboration based on data from DANE.

Although suggestive, this evidence indicates that the reform could have decreased rural school enrollment in the affected departments, consistent with the reform increasing average calendar overlap across affected schools. These plots do not, however, provide causal estimates. Moreover, enrollment is a noisy measure of the overlap channel. As a stock variable, it reflects the cumulative history of prior enrollment and dropout decisions and absorbs scale and composition effects common to all treated schools, such as school mergers, student reshuffling within affected departments, or secular demographic trends in the school-age population driven by changes in fertility or out-migration. Figure 6 indeed shows a negative trend in enrollment across all departments over this period, consistent with such forces operating independently of the reform.

Figure 7 presents the estimates of Equation 4 on dropout rate (Panel a) and grade promotion

rate (Panel b), the flow outcomes that directly capture individual schooling decisions in a given year.³² Two patterns stand out. First, in the years preceding the reform, the estimated coefficients are small and statistically indistinguishable from zero across both outcomes, indicating that schools with different exposure to the calendar change were not on differential trajectories before 2011. Second, following the reform, outcomes deteriorate sharply in schools where the calendar change generated a larger increase in overlap.³³ Dropout rates rise immediately and remain elevated throughout the post-reform period, whereas grade promotion falls persistently, mirroring the dropout response.

Figure 7: Effects of One-Percentage Point Increase in School–Harvest Overlap Induced by the Reform



Notes: The figure plots the estimated coefficients on the interaction between the school-level change in calendar overlap, ΔO_s , and year indicators, relative to the baseline year 2010. Dropout and grade promotion are the shares of students who dropped out or where promoted to the next grade, respectively. Estimates come from the event-study specification Equation 4, which includes school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure). Coefficients are expressed per unit of ΔO_s . The sample includes rural public schools in the affected departments and in the rest of the country, restricted to a balanced panel of schools observed in every year between 2006 and 2018. Vertical bars indicate 90 percent, 95 percent, and 99 percent confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

³²We report the estimates using school enrollment in Appendix Figure A.10 for completeness.

³³One concern is that the reform induced families to relocate across departments. A child who leaves an affected school is recorded there as a transfer rather than a dropout, so relocation does not mechanically enter our dropout measure, but a child who subsequently enrolls in an unaffected school does enter the control observations, and selective departure could alter the composition of treated schools. The evidence indicates that this margin is inactive. Appendix Figure A.11 shows no increase in municipality-level transfer rates after the reform, and no differential trend between affected and unaffected departments in the share of children who migrated within the previous five years.

The static counterpart of the event-study specification, presented in Panel A of Appendix Table A.5, implies that a one-percentage-point increase in the overlap between the school and agricultural calendars, raises the dropout rate by 0.71 to 0.74 percentage points (closely in line with the aggregate cross-country estimates in Section 2) and lowers grade promotion by 0.65 to 0.69 percentage points.³⁴

These estimates capture the effect of increased school–harvest overlap. Dropout rises and grade promotion falls, but the two are not separate responses. Because promotion is measured as a share of enrolled students, its decline largely reflects the rise in dropout rather than an effect on the performance of students who remained in school. When we define promotion conditional on completing the school year as the number of promoted students divided by the sum of promoted and failed students, we find no statistically significant effect of exposure (Appendix Table A.6). The mechanical effect of higher dropout accounts for most of the decline in the unconditional promotion rate. The effects of school–harvest overlap therefore mostly operate through the extensive margin of school continuation, with students in highly exposed schools responding to the higher opportunity cost of schooling by leaving school altogether. This also explains the decline in rural school enrollment among exposed schools in affected departments.

Heterogeneity of the effect: The effects of the reform are concentrated among primary school students. Panels B and C of Appendix Table A.5 present the dose-response estimates separately for primary and secondary school students (Appendix Figure A.12 present the dynamic results). Among primary school students, a one-percentage-point increase in calendar overlap raises the dropout rate by 0.83 percentage points and lowers grade promotion by 0.74 percentage points. Among secondary school students, in contrast, the estimated effects on dropout and grade promotion are close to zero and statistically insignificant.

This pattern is consistent with the opportunity-cost mechanism. School-work concurrence in agriculture tends to be most prevalent among younger children, who are disproportionately

³⁴All estimates are statistically significant at the 1 percent level when clustering at the school level. When clustering at the municipality level, the effects on dropout and grade promotion remain significant at the 1 percent level.

engaged in family subsistence farming (ILO and UNICEF, 2025). Primary school students are therefore the most exposed to agricultural labor demand generated by this type of reforms. The concentration of effects among them is also consistent with the reform accelerating dropout among children at the margin of school continuation, who disproportionately attend primary school.³⁵ We additionally examine heterogeneity by gender in Appendix Figure A.13, and do not find any substantial difference between male and female students.

7.2. Effects of the Reform by Direction of the Change in Overlap

If the effects operate through school–harvest overlap, outcomes should deteriorate where overlap rose and improve where it fell. Table 4 presents the estimates for dropout and grade promotion by direction of the change, using Equation 6.³⁶ The results reveal a clear sign reversal. Among schools where the reform *increased* overlap, a one-percentage-point increase raises dropout by 0.53 percentage points and lowers grade promotion by 0.42 percentage points. Among schools where the reform *reduced* overlap, the effects flip sign. Dropout falls by 1.03 percentage points and grade promotion rises by 1.03 percentage points.

Table 4: Estimates by Direction of the Overlap Change

	Dropout rate		Grade promotion	
	Positive dose (1)	Negative dose (2)	Positive dose (3)	Negative dose (4)
School-Harvest overlap ($ \Delta O_s \times \text{Post}$)	0.531*** (0.096)	-1.031*** (0.181)	-0.420** (0.178)	1.032*** (0.241)
Observations	231,855	225,368	231,855	225,368
Baseline controls $\times$ year FE	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Notes: The table reports static estimates of Equation 6. The treatment variable is the absolute value of the change in overlap, so coefficients are expressed per unit of $|\Delta O_s|$ in each group's own direction. A negative coefficient in the negative-dose columns indicates that the outcome falls as overlap decreases. All specifications include school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics, on the balanced panel 2006–2018. Standard errors clustered at the school level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Own elaboration based on data from DANE.

³⁵Section 8 shows that the reform also raised the probability that children aged 10 to 15 work in agriculture, with the increase concentrated among children with completed primary and incomplete secondary education.

³⁶The dynamic results are presented in Appendix Figure A.14, where we do not see evidence of existing pre-trends by subgroups.

These results provide strong evidence for our mechanism. Schools more exposed to calendar overlap face a deterioration on schooling outcomes, and schools where overlap fell see an improvement. This sign reversal is difficult to reconcile with any confounder common to the affected departments, and indicates that the overlap between schooling and agricultural labor demand, rather than the calendar change per se, drives the effects on school progression.

7.3. Robustness of the School-Level Estimates

We provide additional evidence supporting our main results through four complementary robustness exercises. First, we show that our findings are not sensitive to alternative definitions of harvesting intensity months (i.e., the set defined in $\mathcal{I}_j$). Second, we re-estimate the main effects using alternative estimators that account for potential heterogeneity induced by the continuous treatment, combined with progressively more local control groups, yielding 21 estimator-by-control-group combinations per outcome. Third, we add treated-by-year fixed effects, which absorb any shock common to schools in the affected departments. Fourth, we complement the dose-response design with an intent-to-treat estimate using the SDID procedure. We provide complementary details about all robustness checks in Appendix D.

Definition of Harvesting Intensity. Our overlap measure in Equation 2 defines harvesting-intensive months as the top quartile of the annual distribution of harvesting activity, corresponding to the three most intensive months of the year. To verify that the results do not depend on this threshold, we re-estimate our main specification using stricter definitions of the dose, constructing ΔO_s from only the single most intensive month and from the two most intensive months. We report the resulting estimates in Appendix Figure D.1.

The effects of calendar overlap on school progression are robust across definitions of harvesting intensity. Dropout rises and grade promotion falls in schools where overlap with the most intensive harvesting periods increased, with estimates that retain their sign, magnitude, and statistical significance under all three definitions. The robustness of the results under progressively

stricter definitions indicates that the effects on school progression are not driven by the specific threshold used to define it.

Alternative Estimators. Our dose-response design assigns a continuous treatment intensity to every school. However, two-way fixed effects estimators with continuous treatments can place negative weights on some treatment effects, potentially yielding estimates that are not a convex combination of the underlying school-level effects (de Chaisemartin and D’Haultfœuille, 2020; Callaway and Sant’Anna, 2021). To address this, we re-estimate the dose-response specification using three alternative estimators. The first is the two-way fixed effects specification described in Section 6. The second and third are the normalized and non-normalized versions of the heterogeneity-robust estimator of de Chaisemartin et al. (2024), which expresses the effect per unit of treatment dose.

A second concern is that schools in distant regions of Colombia may provide a poor counterfactual for schools in the affected departments, either because they face different agro-climatic conditions or because they are exposed to different regional shocks. We therefore re-estimate all specifications restricting the control group to progressively more local comparisons, starting with schools in the six departments bordering the affected ones and then narrowing to schools located within 40, 45, 50, 55, and 60 kilometers of the boundary of the affected departments.³⁷ These restrictions trade sample size for comparability, as nearby schools are more likely to share climatic conditions, crop portfolios, and exposure to local economic shocks.

Appendix D.2 summarizes the full robustness battery in specification-curve form. For each outcome, we plot the 21 point estimates generated by crossing the three estimators with the seven control groups, together with 95 percent confidence intervals clustered at both the school and the municipality level. Each specification is colored according to the p-value of a joint pre-trend test. Red indicates the null of parallel pre-trends is not rejected at the 10 percent level, blue indicates rejection at the 10 but not the 5 percent level, and dark blue indicates rejection at the 5 percent

³⁷The narrowest band—40 kilometers—approximately corresponds to the median distance from the boundary among control schools in the bordering departments (41 kilometers). See Appendix D for details.

level. We present our main estimate, which corresponds to the one presented in Section 7 in green color. The lower panel of each curve identifies the estimator and control group, allowing readers to verify that no single modeling choice drives the results.

The curves show that the estimated effects are stable in sign and magnitude across the entire specification space. Each of the 21 estimates on dropout is positive and statistically significant, ranging from 0.59 to 0.90. The point estimates on grade promotion are uniformly negative. No dropout specification rejects the null of parallel pre-trends at the 10 percent level, and only two of the 21 grade promotion specifications do.³⁸

Common Shocks in Affected Departments. The reform created a one-off academic break during the transition year, which De Roux and Riehl (2022) study in the context of college enrollment. Any shock of this kind, common to all public schools in Nariño and Valle del Cauca, is not excluded from the dose coefficient, since the dose is nonzero only for affected schools. We therefore re-estimate the main specification adding a full set of treated-by-year fixed effects, which absorb any shock common to affected schools. The estimates for dropout and grade promotion are essentially unchanged. Appendix D.3 reports the results. The split estimates of Section 7.2 provide a second check for the same concern, since no common shock can move outcomes in opposite directions across schools whose overlap rose and fell.

Intent-to-Treat Estimates. Finally, we complement the dose-response evidence with estimates of the reform’s effect using the synthetic difference-in-differences estimator of Arkhangelsky et al. (2021), which compares schools in affected departments with a data-driven weighted combination of schools in unaffected ones. Whereas the dose-response design identifies the effect of the overlap channel from within-treatment variation in exposure, the SDID estimates capture the intent-to-treat effect of the reform, including any component common to all schools in affected departments. We present the results in Appendix D.4. The reform raised dropout rates by approximately 1.7

³⁸For completeness, we also report robustness checks for school enrollment, even though, as explained in Section 7.1, this outcome is a stock, measured in logs, and absorbs scale, composition, and demographic effects. The estimates are persistently negative and we fail to reject the null of parallel trends in every case.

percentage points on average and lowered grade promotion by approximately 2 percentage points. These aggregate estimates exceed the dose-response estimates evaluated at the average overlap change because the two designs recover different estimands. SDID captures the full effect of the reform on affected departments, including components common to all treated schools regardless of their overlap change, whereas the dose-response design isolates the component that varies with exposure to the overlap channel.

8. Children’s Agricultural Work and the Reform

The deterioration of schooling outcomes in regions where school–harvest overlap increased is consistent with a time-allocation mechanism in which children substitute schooling for agricultural work. We examine this directly by testing whether the calendar reform increased children’s agricultural work, using nationally representative household survey data of Colombia and a municipality-level difference-in-differences design. Throughout this section, children’s agricultural work refers to a single outcome, an indicator equal to one if a child aged 10 to 15 reports working in agriculture or silviculture, constructed from the household survey’s employment module.³⁹ We estimate the following specification:

$$E_{i,m \in d,t} = \sum_{k \neq 2010} \tau_k (W_m \times \mathbb{1}[t = k]) + \alpha_m + \beta_t + \sum_k (\mathbb{1}[t = k] \times X_{m,t_0})' \gamma_k + \epsilon_{imt}, \quad (7)$$

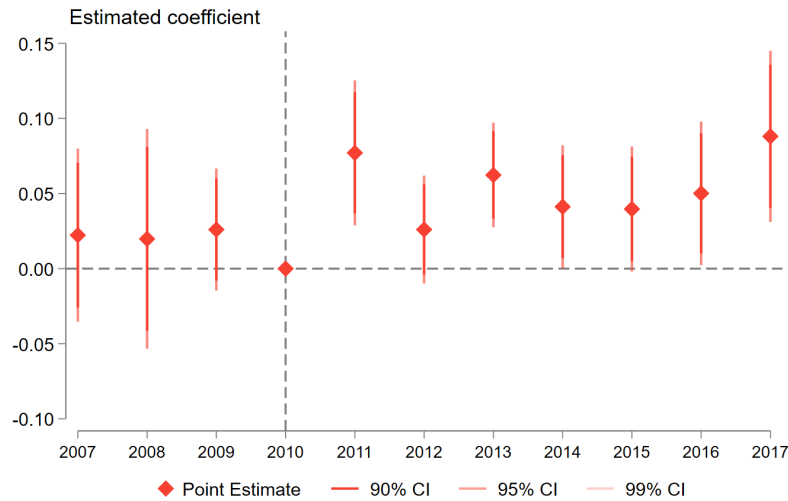
where $E_{i,m \in d,t}$ is the children’s agricultural work indicator defined above for child i in municipality m of department d in year t . W_m is a treatment indicator equal to one if municipality m belongs to one of the affected departments. As control municipalities, we use a matched set of unaffected

³⁹The survey records employment only from age 10, so the analysis covers the upper end of primary school and the first years of secondary school, and cannot speak to the work of younger children.

departments comparable to the treated ones, namely Risaralda and Sucre.⁴⁰ The terms α_m and β_t denote municipality and year fixed effects, while X_{m,t_0} is a vector of predetermined municipality-level characteristics, interacted with year fixed effects to flexibly control for time-varying shocks that may be correlated with both municipality characteristics and child labor outcomes.⁴¹

Figure 8 reports the estimated τ_t coefficients. Prior to the reform, the probability of children engaging in agricultural work evolves in parallel across affected and unaffected municipalities, with coefficients close to zero, supporting the parallel-trends assumption underlying the design. Following the reform, the probability rises in affected municipalities and remains elevated through the end of the sample period. The persistence of the effect matches the dynamics of the dropout response documented in Section 7 and is consistent with a lasting reallocation of children's time toward farm work in affected areas.

Figure 8: Effect of the Reform on Children's Agricultural Work



Notes. This figure plots the estimated coefficients τ_t from Equation 7. Source: Own elaboration based on data from DANE.

Appendix Table A.7 presents the static counterpart of the event study estimates. The reform

⁴⁰To select the comparison group, we first restrict the pool to unaffected departments whose primary agricultural products overlap with those of the affected departments (stimulants and rice), and then select the closest departments based on a standardized index of pre-treatment covariates, including the share of agriculture in departmental GDP, the log of the rural population, the log of public expenditure, and the log of public revenues. This procedure identifies Risaralda, Sucre, and Arauca. Arauca is excluded because it is not surveyed at the same frequency as the rest of the Colombian departments in the household survey.

⁴¹The pre-treatment controls include per capita GDP, the number of terrorist attacks at the municipality level, the homicide rate, rural population, and agricultural GDP.

raised the probability of agricultural work among children aged 10 to 15 by 3.9 percentage points, a 56 percent increase over a pre-reform baseline of 7 percent in affected departments. Among children aged 10 to 12 the increase is 3.0 percentage points over a baseline of 3.6 percent, or 83 percent, while among those aged 13 to 15 it is 4.9 percentage points over a baseline of 10.4 percent, or 47 percent. The point estimate is larger in absolute terms among older children, who are more likely to be employed at baseline, although the difference between the two age groups is not statistically significant ($p = 0.36$). Relative to their own pre-reform mean, the response is larger among 10-to-12 year olds, the group closest to the primary grades where the schooling effects concentrate.

Panel B of Appendix Table A.7 splits children by current school enrollment. Among children who report attending school, the reform raised the probability of agricultural work by 3.8 percentage points, an 87 percent increase over a pre-reform baseline of 4.4 percent. Among children not attending, the increase is 8.9 percentage points over a baseline of 26 percent, or 34 percent, and is estimated less precisely given the smaller group. The response among enrolled children indicates that the trade-off between schooling and harvest work operates while children are still in school, rather than only after they leave. Enrollment is itself an outcome of the reform, so the enrolled subsample changes composition after 2010, and the split should be read as describing where the work response is located rather than as a conditional treatment effect.

These estimates are reinforced by descriptive patterns in the same data. Appendix Figure A.15 provides three complementary views of children's agricultural work. Panel A.15a plots the share of children working in agriculture over time and shows that, in affected departments, this share approximately doubled between the pre-reform period and 2017, while it remained essentially flat in the rest of the country. Panel A.15b summarizes this pattern as a raw before-after comparison. The share of children with at most completed primary education working in agriculture rose by approximately one percentage point in affected departments (about a 20 percent increase relative to pre-reform levels) and declined slightly elsewhere. Panel A.15c disaggregates the post-reform increase by educational attainment within affected departments and shows that the

rise is concentrated among children with completed primary education and incomplete secondary education, which are precisely the margins along which dropout increased due to the reforms.

Taken together, the calendar reform increased the probability that rural children work and reduced the share who remained in school. The event studies cannot establish the order in which these responses occur. Work may rise because children are drawn into the harvest while still enrolled, or because children who leave school then take up farm work, and the two are not mutually exclusive. What the evidence establishes is a reallocation of children's time toward agricultural work that accompanies the deterioration in schooling, which is what a time-allocation mechanism predicts. The event-study coefficients in Figures 7a and 8 both rise from 2011, and promotion conditional on completing the school year does not respond to exposure (Appendix Table A.6), so we find no evidence that the reform operated through academic difficulty among students who remained in school. Overall, greater overlap between the school year and the local harvest appears to raise the opportunity cost of schooling, leading households to substitute children's schooling with labor supply during peak agricultural periods.

9. Discussion and Conclusion

This paper studies the economic consequences of conflicts between timing institutions and local economic activity. We focus on school calendars in agrarian economies, where time for schooling can overlap with seasonal peaks in agricultural work. Our findings show that these timing conflicts can meaningfully disrupt human capital accumulation by increasing school dropout.

Our evidence highlights two distinct institutional origins of the same timing conflict. The first is historical persistence. Using newly assembled cross-country data on school and harvest calendars, we show that school calendars in many former colonies remain closely related to those of their colonizers, particularly where colonial rule continued after the colonizer's public education system had been institutionalized. Among former colonies with greater calendar

similarity to their colonizer, school–harvest overlap increases sharply as local harvest timing moves farther from the inherited vacation period. Instrumental-variable estimates based on colonial calendar timing imply economically meaningful schooling costs: greater school–harvest overlap substantially increases the share of primary-school-age children who are out of school.

The second source is domestic standardization. Colombia developed its school calendars after independence, making its experience distinct from the colonial-inheritance channel. Beginning in 2011, public schools in Nariño and Valle del Cauca shifted from a regionally differentiated calendar to the calendar used in most of the country. Because local crop cycles differ across locations, the same national standard increased school–harvest overlap in some schools and reduced it in others. Where overlap rises, dropout increases and grade promotion falls. Where it falls, both improve. The estimates are robust to alternative measures of harvesting intensity, to alternative estimators and clustering choices, to progressively more local control groups, and to absorbing shocks common to all affected schools.

The Colombian evidence also illuminates the household response underlying these schooling effects. Nationally representative household data show that agricultural work among children aged 10–15 significantly rises in affected municipalities, including among children who remain enrolled. This pattern is consistent with a time-allocation mechanism in which resource-constrained agricultural households rely on children’s labor when school and harvest demands coincide. The evidence does not establish whether work precedes eventual school exit, but it shows that deteriorating schooling outcomes are accompanied by a substantial reallocation of children’s time toward farm work.

The cross-country and Colombian designs are therefore complementary rather than interchangeable. The former identifies variation in timing conflict associated with historical institutional inheritance, while the latter studies a domestically generated change in timing within a common national institutional environment. Their outcomes also differ. The cross-country measure is a national stock of children out of school, whereas the Colombian outcome is an annual flow of dropout among previously enrolled students. Both designs nonetheless point

in the same direction and yield effects of a similar magnitude. This convergence across distinct populations, outcomes, and sources of variation suggests that timing institutions, captured by the school–harvest conflict, are an economically important margin of human capital accumulation rather than an artifact of either empirical design.

Our findings do not imply that school calendars should always be decentralized or mechanically synchronized with local harvests. Standardization can generate valuable system-wide benefits, including administrative coordination, teacher mobility, common assessment schedules, and smoother transitions across schools and into higher education. Our design does not measure these benefits and therefore cannot determine the net welfare effect of calendar unification. The pre-reform calendars were themselves poorly matched to local agriculture in many places, and the national calendar reduced overlap for roughly one-third of affected schools. Therefore, the relevant policy question is not simply whether calendars should be unified or decentralized, but how institutional standardization should account for predictable local variation in economic activity.

This broader trade-off extends beyond school calendars. Many institutions coordinate activity by determining when actions take place, including school terms, examinations, fiscal years, elections, work schedules, and administrative deadlines. Standardization can generate coordination gains, but common timing rules may also create costs when economic activity varies systematically across locations or seasons. Our findings therefore point to a broader principle: institutional design concerns not only *what* rules require, but also *when* they require it. Understanding how to balance local temporal alignment with system-wide coordination is an important direction for future research.

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Appendix

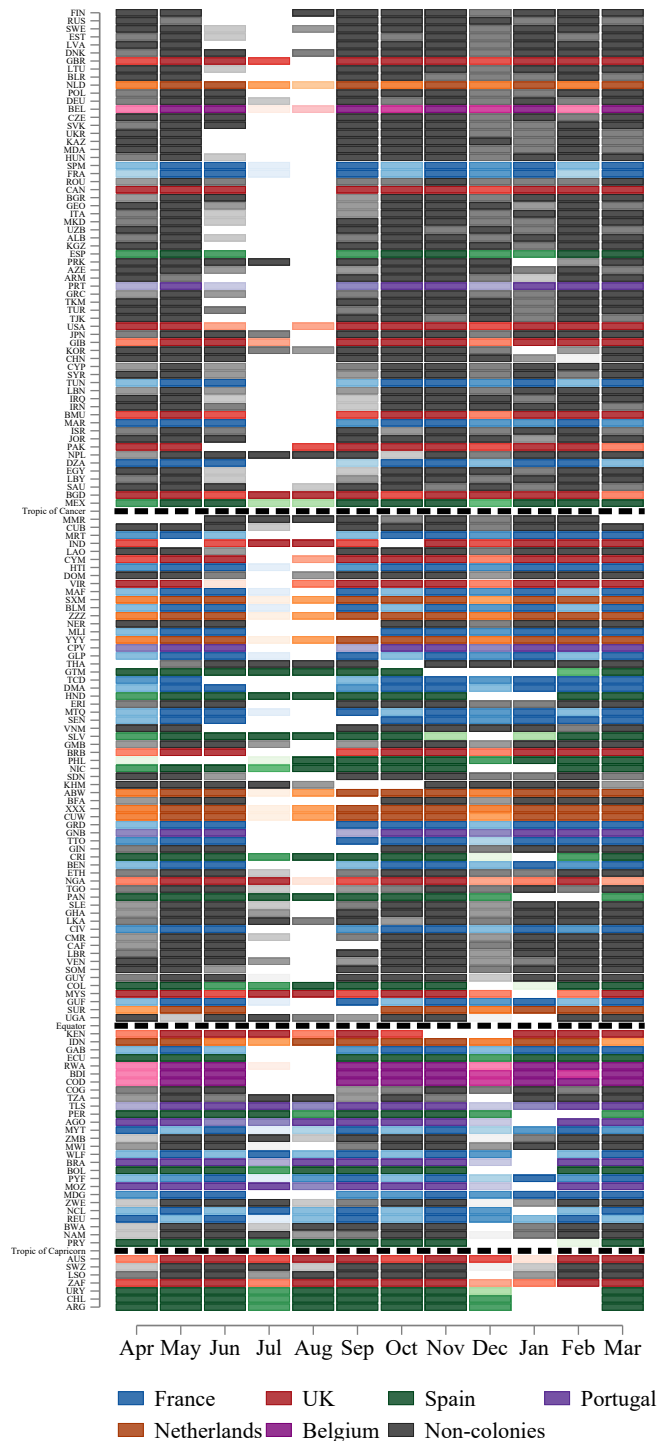
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A. Additional Figures and Tables

A.1. Figures

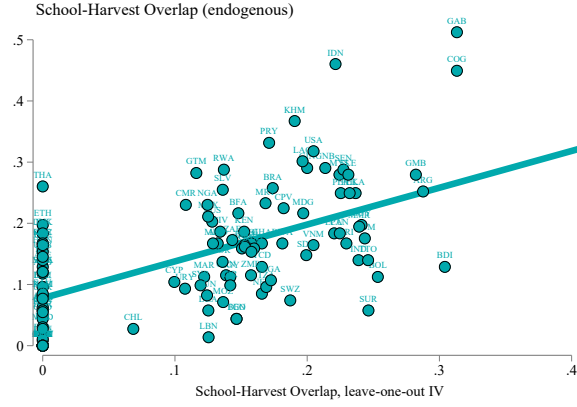
Appendix Figure A.1: Worldwide School Calendars by Latitude and Colonizer



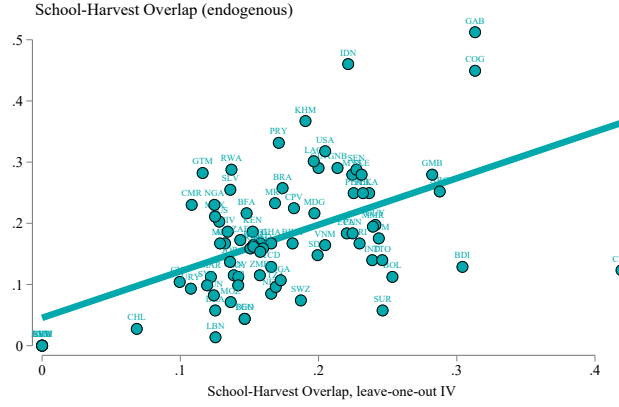
Notes: The figure plots the monthly share of annual school days across countries, ordered vertically by geographic latitude. Dashed horizontal lines indicate key latitude benchmarks (Tropic of Cancer, Equator, and Tropic of Capricorn). Bars are color-coded by historical colonizing power (or non-colony status). For each country, the information corresponds to the most recent school calendar data available. Source: Own elaboration based on official sources.

Appendix Figure A.2: First-Stage Relationship

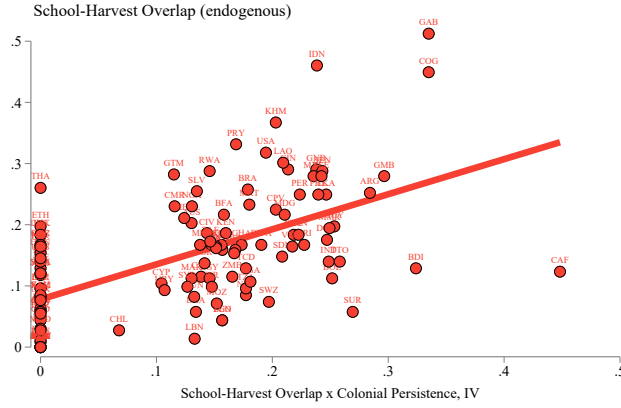
(a) Leave-one-out overlap, Full sample



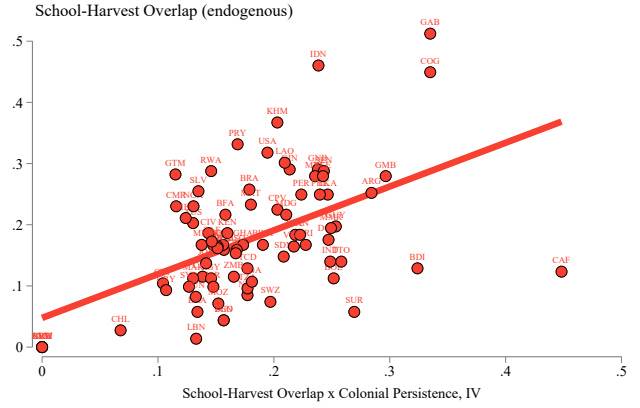
(b) Leave-one-out overlap, Colonies



(c) Overlap weighted by colonial exposure, Full sample

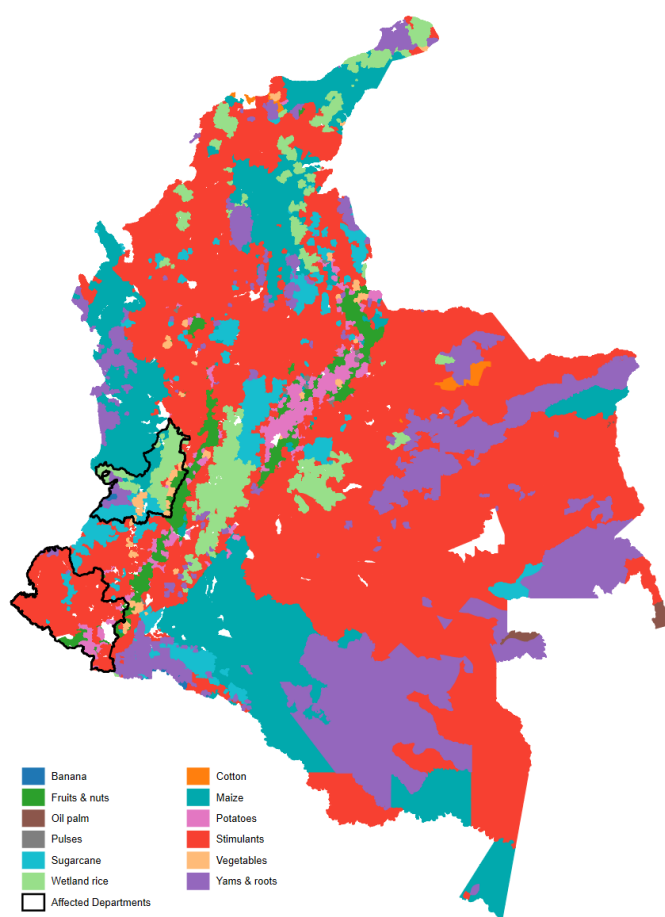


(d) Overlap weighted by colonial exposure, Colonies

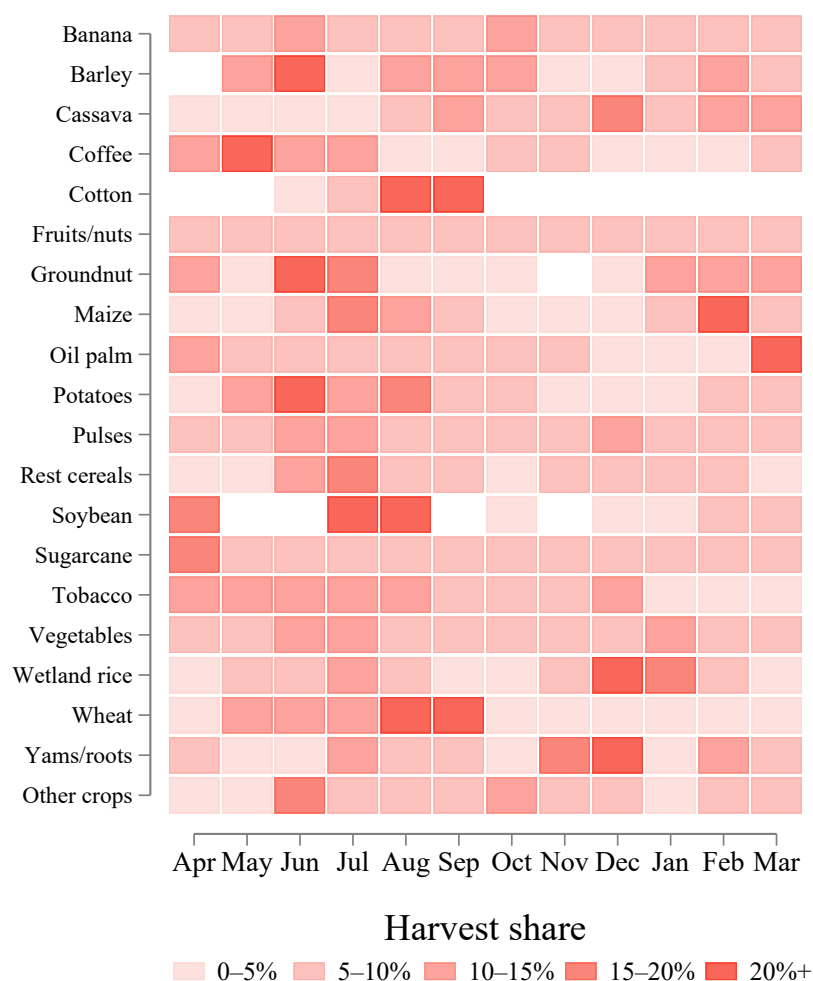


Notes: Each panel plots the observed overlap between school and harvesting calendars, O_i , against the instrument, with each dot representing a country and the solid line a linear fit. Panels **a** and **b** use the baseline instrument Z_i , for the full sample and for former colonies. Panels **c** and **d** use the overlap weighted by colonial exposure instrument Z_i^C . The figures display the unconditional relationship between the observed overlap and the instruments. Both instruments are expressed as shares of the year, on the same scale as the overlap measure. Source: Own elaboration based on data described in Appendix B.1.

Appendix Figure A.3: Main Crop by Vereda

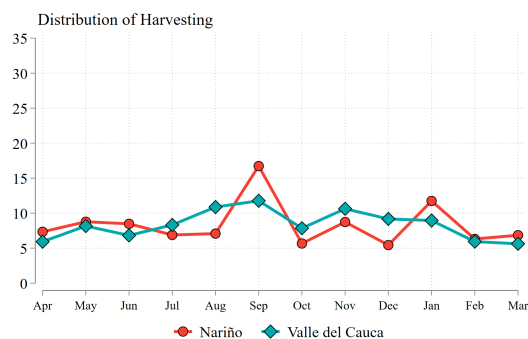


Notes. The figure shows the main harvested crop in each *vereda*. Stimulants include coffee, tea, and cocoa. *Veredas* shown in white indicate areas for which no information is available. Source: Own elaboration based on data from the Global Agro-Ecological Zones (GAEZ) v4 - FAO.

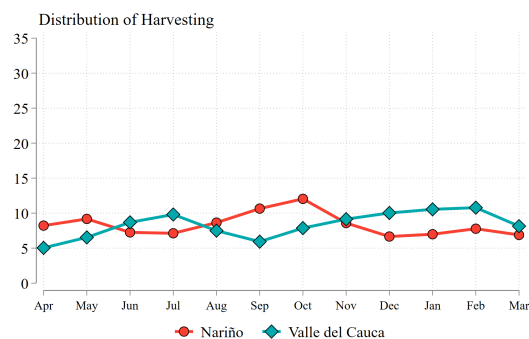
Appendix Figure A.4: Harvesting Calendars in Affected Departments

Notes. This figure plots the proportion of days dedicated to harvesting activities in each month, relative to the total annual harvesting for each crop, averaged across Nariño and Valle del Cauca. Crops are ordered alphabetically. Darker cells indicate larger shares, grouped in bins of 0-5, 5-10, 10-15, 15-20, and above 20 percent. White cells indicate months with no harvesting. Coffee refers to the stimulants category, which also includes tea and cocoa. Data come from the Colombian Ministry of Agriculture.

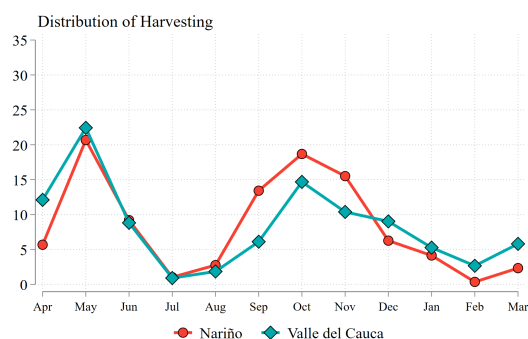
Appendix Figure A.5: Harvesting Calendars - Nariño and Valle del Cauca



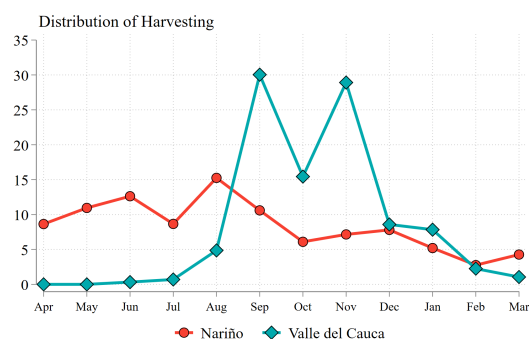
(a) Banana



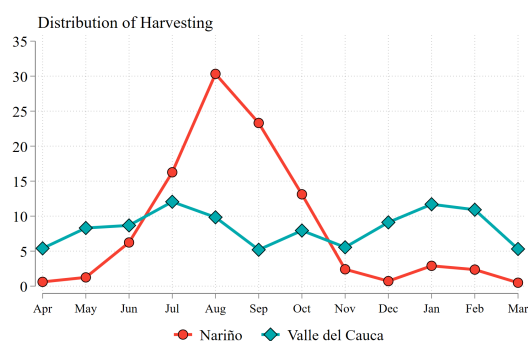
(b) Fruits and Nuts



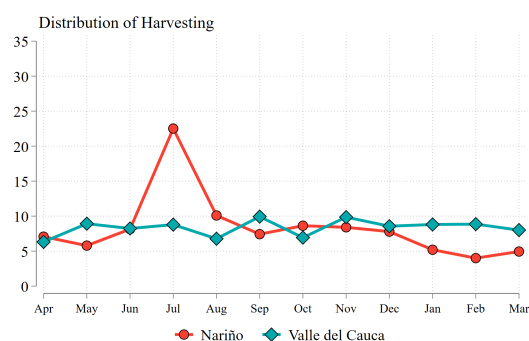
(c) Maize



(d) Potato and Sweet Potato



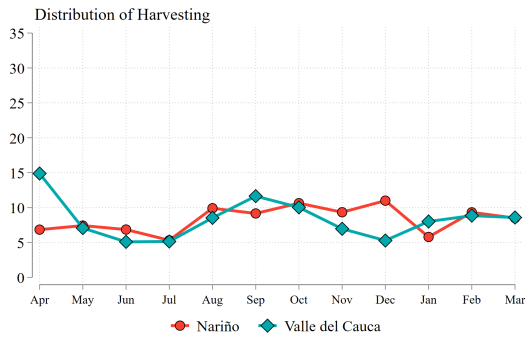
(e) Stimulants



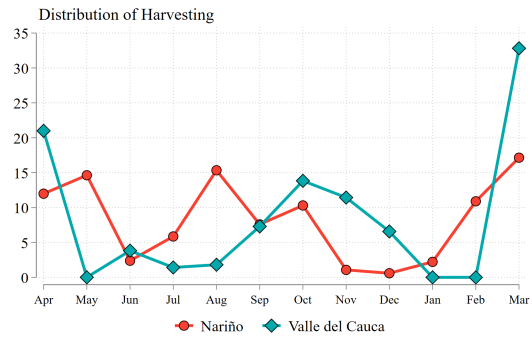
(f) Sugarcane

Notes. The figures show the proportion of days dedicated to harvesting activities in each month, relative to the total annual harvesting for each crop. Source: Own elaboration based on data from the Colombian Ministry of Agriculture.

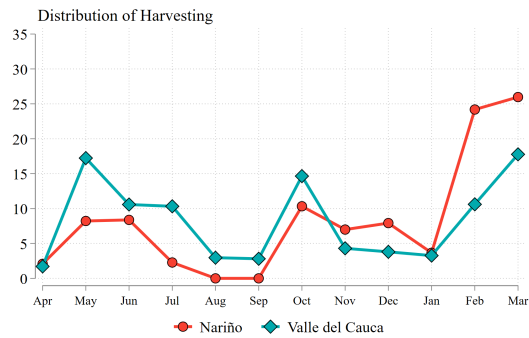
Appendix Figure A.6: Harvesting Calendars - Nariño and Valle del Cauca



(a) Vegetables



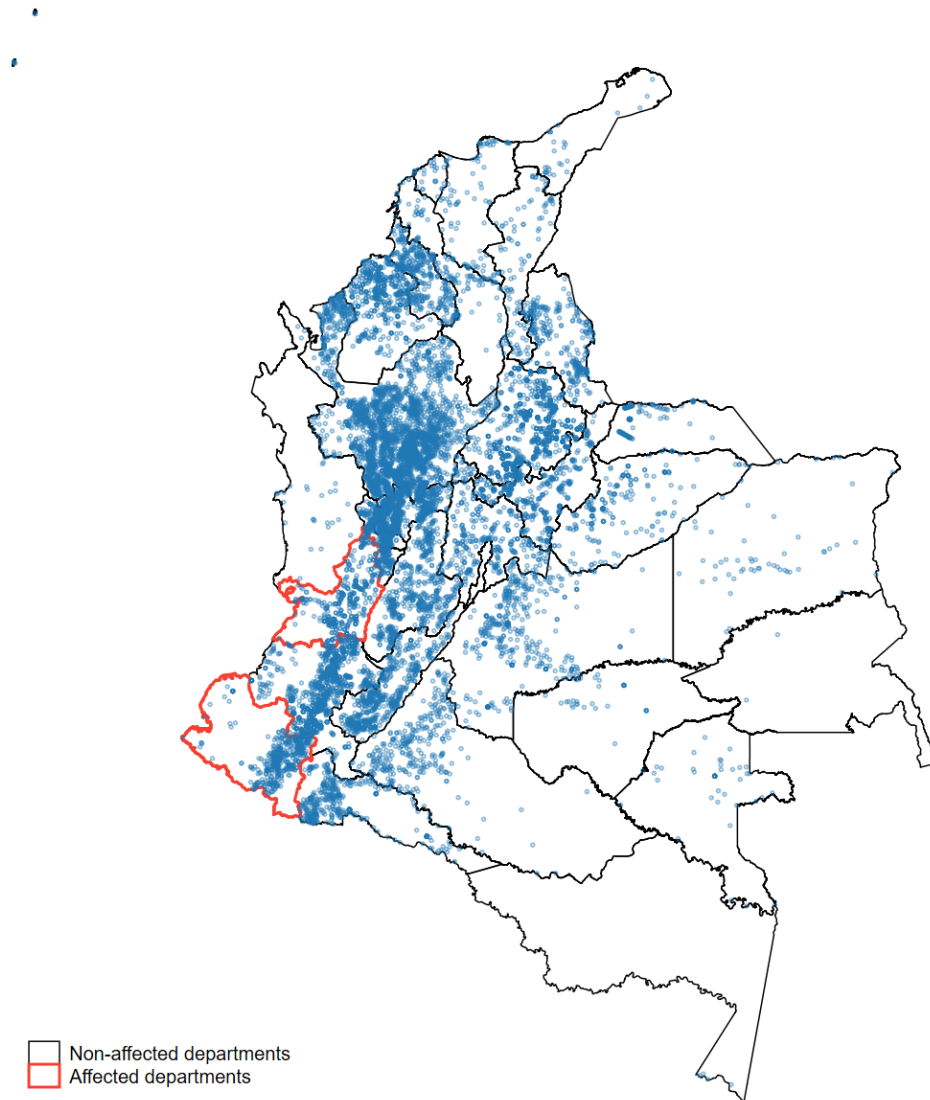
(b) Wetland Rice



(c) Yams and Other Roots

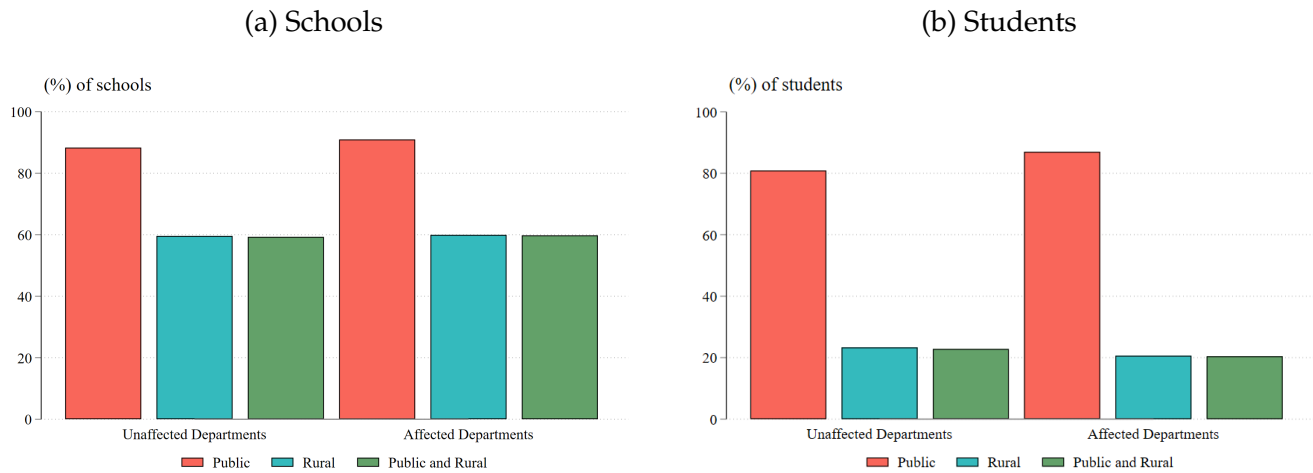
Notes. The figures show the proportion of days dedicated to harvesting activities in each month, relative to the total annual harvesting for each crop. Source: Own elaboration based on data from the Colombian Ministry of Agriculture.

Appendix Figure A.7: Spatial Distribution of Schools in Affected and Unaffected Departments



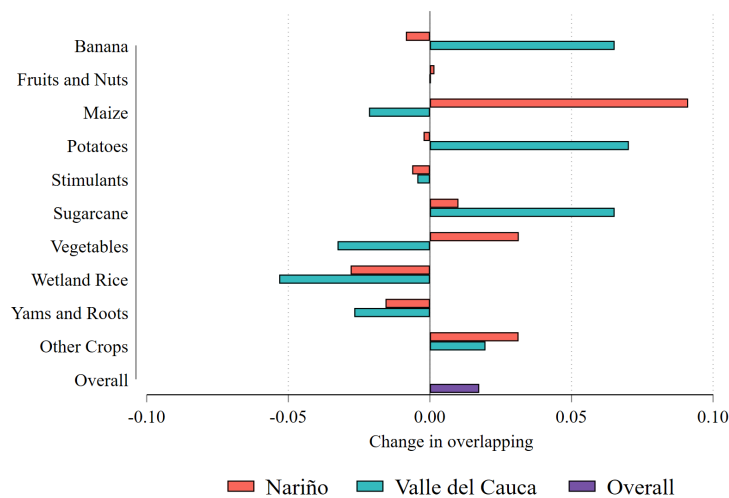
Notes. Each dot represents a school in the estimation sample. Source: Own elaboration based on data from DANE.

Appendix Figure A.8: Proportion of Public and Rural Schools and Students



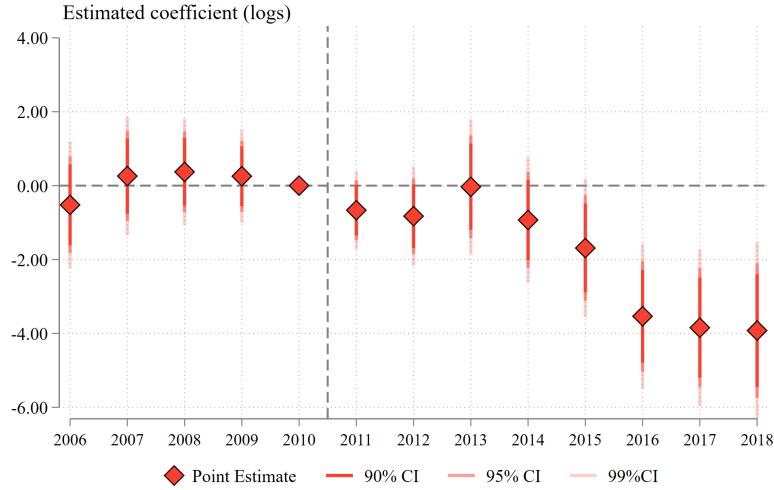
Notes: The figure shows the percentage of schools that are public, rural, or both in 2010, in affected departments (i.e., Nariño and Valle del Cauca), and in the rest of the country. Data come from the Colombian census of schools (Form C-600).

Appendix Figure A.9: Change in Calendar Overlap Induced by the Reform



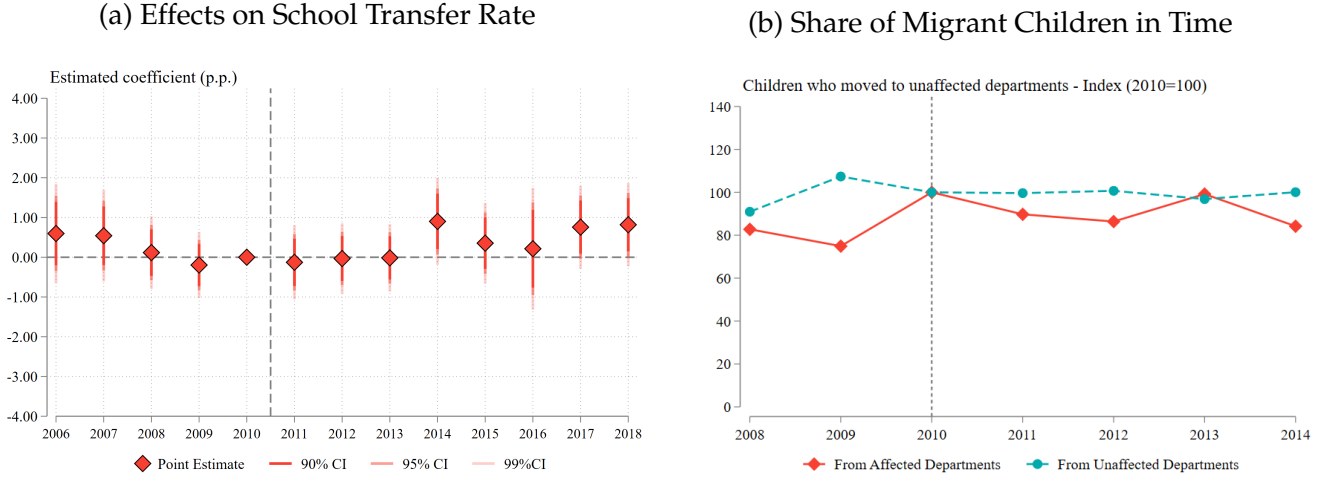
Notes: The figure plots the average change in overlap between the academic and harvesting calendars before and after the reform, by crop, based on the most harvested crop in each *vereda* where schools are located. The overall average, shown by the purple bar, equals 0.017.

Appendix Figure A.10: Effect of the Reform on School Enrollment



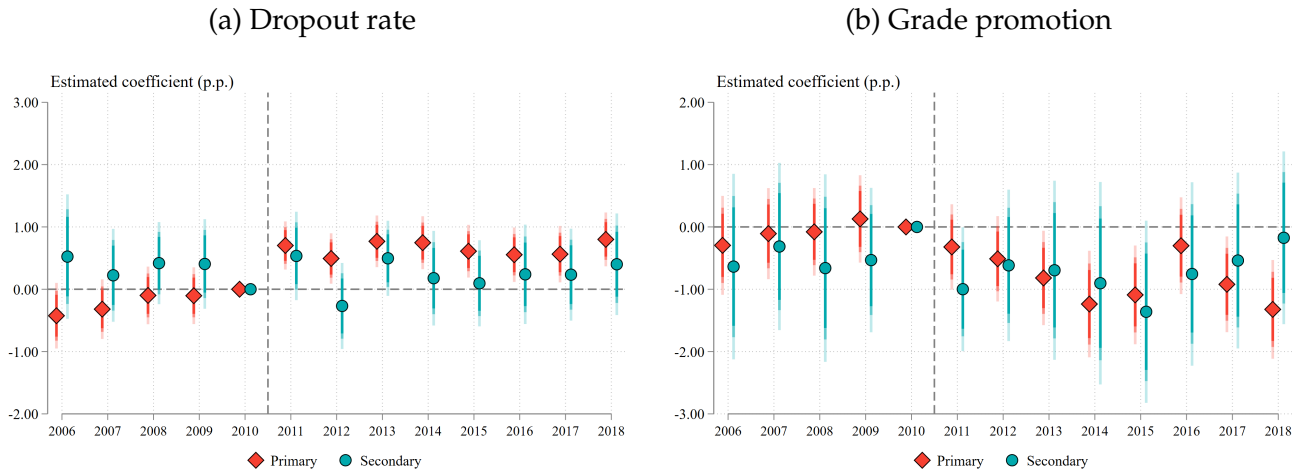
Notes: The figure plots the estimated coefficients on the interaction between the school-level change in calendar overlap, ΔO_s , and year indicators, relative to the baseline year 2010, for school enrollment, measured as the log of the total number of students per school. Estimates come from the event-study specification Equation 4, which includes school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure). Coefficients are expressed per unit of ΔO_s . The sample includes rural public schools in the affected departments and in the rest of the country, restricted to a balanced panel of schools observed in every year between 2006 and 2018. Vertical bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

Appendix Figure A.11: No Evidence of Reform-Induced Cross-Department Migration



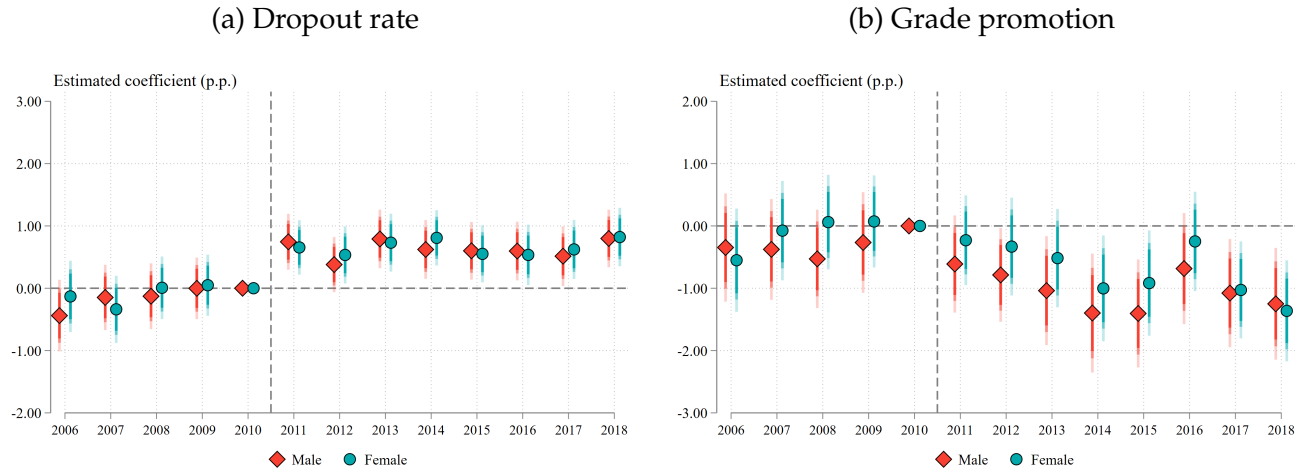
Notes: Panel A.11a plots the estimated coefficients on the interaction between the municipality-level change in calendar overlap and year indicators, relative to the baseline year 2010, for the transfer rate of rural public schools aggregated to the municipality level. The municipal dose is the baseline-enrollment-weighted average of the school-level change in overlap, ΔO_s . The specification includes municipality and year fixed effects and interactions between year fixed effects and baseline municipality characteristics, estimated on the balanced all-country panel of municipalities between 2006 and 2018. Coefficients are expressed per unit of the dose. Vertical bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the municipality level. Source: Own elaboration based on data from DANE. Panel (b) plots, among children aged 5 to 15 interviewed in departments other than Nariño and Valle del Cauca, the share who resided in an affected department five years before the interview, using the GEIH migration module for survey years 2013 to 2019. Each observation is dated to the residence reference year, five years before the interview, and the series is indexed to 100 in 2010. Source: Own elaboration based on data from DANE.

Appendix Figure A.12: Heterogeneous Effects of the Reform by Educational Level



Notes: Each panel plots the estimated coefficients on the interaction between the school-level change in calendar overlap, ΔO_s , and year indicators, relative to the baseline year 2010, from the event-study specification of Equation 4, estimated separately for the outcomes of primary and secondary students. Each estimation includes school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure), and uses the balanced all-country panel of rural public schools for which the group-specific outcome is observed in every year between 2006 and 2018. Coefficients are expressed per unit of ΔO_s . Vertical bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

Appendix Figure A.13: Heterogeneous Effects of the Reform by Gender

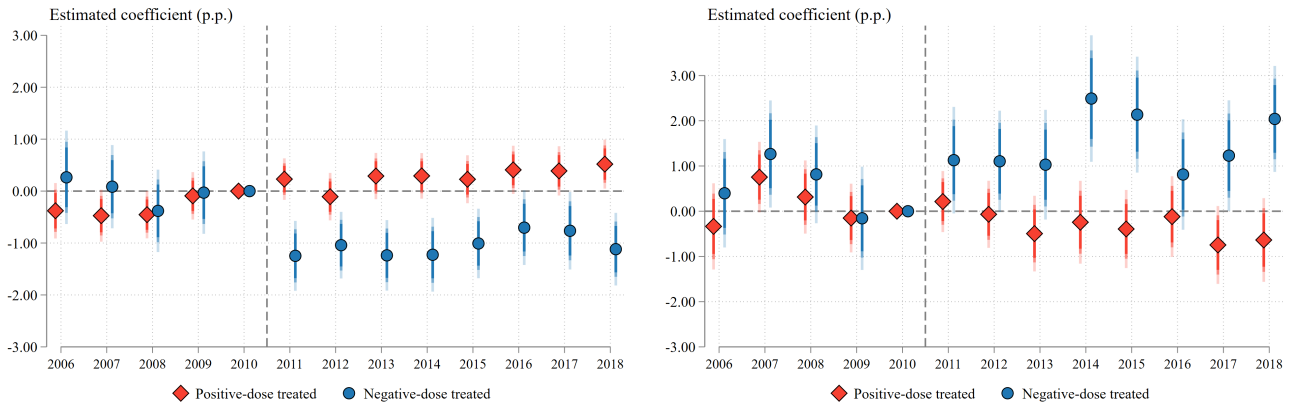


Notes: Each panel plots the estimated coefficients on the interaction between the school-level change in calendar overlap, ΔO_s , and year indicators, relative to the baseline year 2010, from the event-study specification of Equation 4, estimated separately for the outcomes of male and female students. Each estimation includes school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure), and uses the balanced all-country panel of rural public schools for which the group-specific outcome is observed in every year between 2006 and 2018. Coefficients are expressed per unit of ΔO_s . Vertical bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

Appendix Figure A.14: Effects of the Reform by Direction of the Change in Overlap

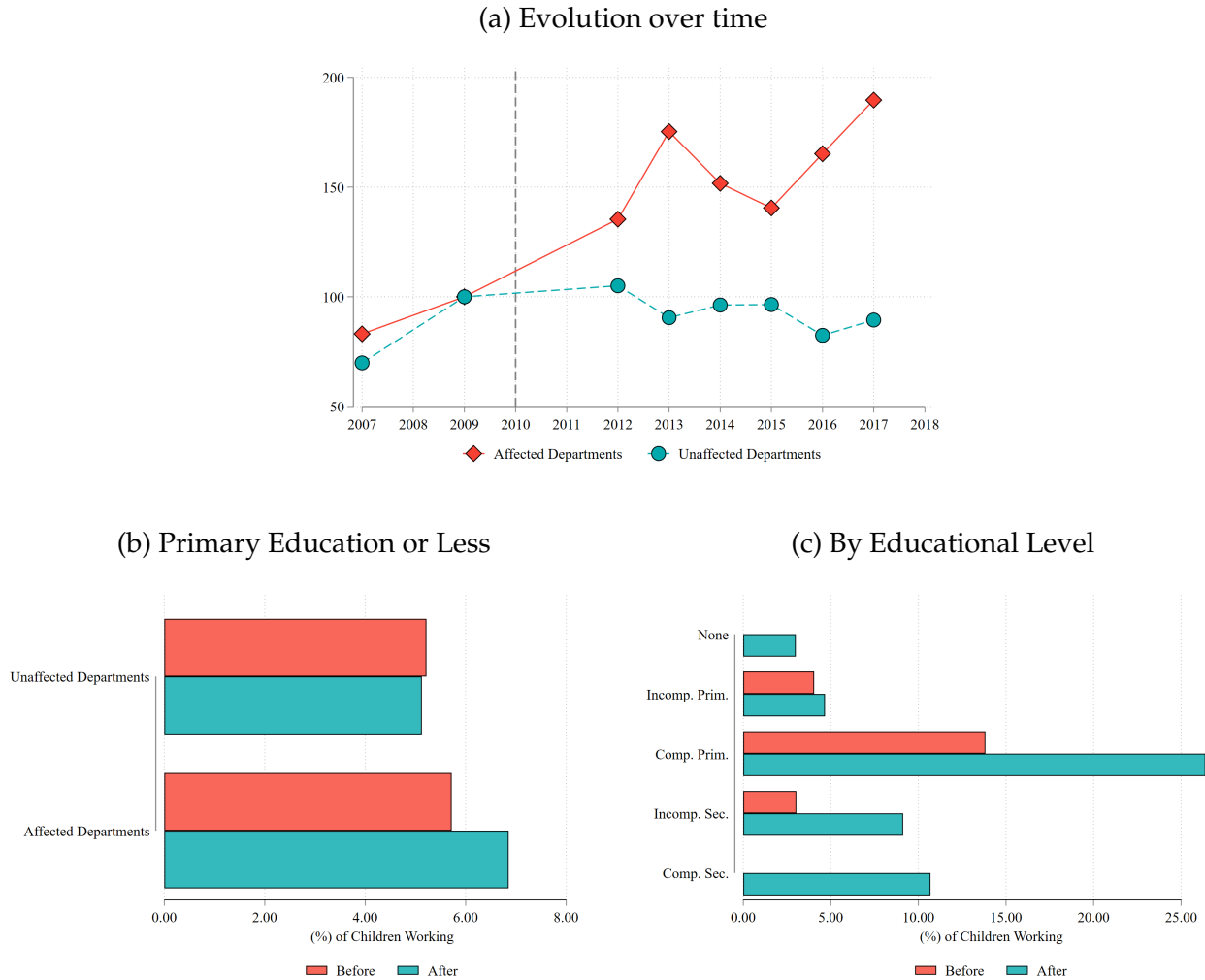
(a) Dropout rate

(b) Grade promotion



Notes: Each panel plots the estimated dynamic effects of the reform per unit of $|\Delta O_s|$, the absolute value of the school-level change in calendar overlap, interacted with year indicators relative to the baseline year 2010, estimated separately for treated schools whose overlap increased (positive dose) and treated schools whose overlap decreased (negative dose), each compared against the full all-country control group. Coefficients are therefore read in each group's own direction. For the negative-dose group, a negative coefficient indicates that the outcome falls as overlap decreases. Dropout and grade promotion are the shares of students who dropped out or were promoted to the next grade, respectively. Estimates come from the event-study specification of Equation 4, which includes school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure). The sample includes rural public schools in the affected departments and in the rest of the country, restricted to a balanced panel of schools observed in every year between 2006 and 2018. Vertical bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

Appendix Figure A.15: Share of Children Working in Agriculture



Panel (a) shows the evolution of the share of children working in agriculture and silviculture over time, normalized to 100 in 2009, for Nariño and Valle del Cauca compared to the rest of Colombia. Panel (b) shows the share of children with incomplete or complete primary education working in agriculture and silviculture before and after the academic calendar change, for Nariño and Valle del Cauca compared to the rest of Colombia. Panel (c) shows the share of children working in agriculture and silviculture before and after the academic calendar change, by educational level, for Nariño and Valle del Cauca. Source: Own elaboration based on data from DANE.

A.2. Tables

Appendix Table A.1: Descriptive Statistics of Key Variables

	Mean	SD	p25	p50	p75	Min	Max
<i>Panel A: All countries</i>							
O_i (school–harvest overlap)	0.148	0.103	0.075	0.142	0.200	0.000	0.512
Out-of-school share	0.089	0.106	0.016	0.042	0.122	0.000	0.451
Z_i (instrument)	0.117	0.100	0.000	0.136	0.194	0.000	0.419
Z_i^C (exposure-weighted instrument)	0.123	0.105	0.000	0.143	0.203	0.000	0.448
Colony indicator	0.675	0.470	0.000	1.000	1.000	0.000	1.000
<i>Panel B: Colonies (colony = 1)</i>							
O_i (school–harvest overlap)	0.178	0.103	0.112	0.167	0.249	0.000	0.512
Out-of-school share	0.115	0.117	0.022	0.070	0.176	0.002	0.451
Z_i (instrument)	0.174	0.071	0.136	0.166	0.224	0.000	0.419
Z_i^C (exposure-weighted instrument)	0.182	0.075	0.142	0.173	0.228	0.000	0.448
<i>Panel C: Non-colonies (colony = 0)</i>							
O_i (school–harvest overlap)	0.087	0.069	0.027	0.082	0.145	0.000	0.260
Out-of-school share	0.033	0.042	0.009	0.019	0.042	0.000	0.202
Z_i (instrument)	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Z_i^C (exposure-weighted instrument)	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Notes. Z_i is the baseline instrument and Z_i^C the exposure-weighted instrument defined in Appendix B.3. O_i is the observed overlap between the school calendar and the agricultural calendar. Out-of-school share is the share of primary-school-age children out of school. Colony indicator equals 1 if the country was a former colony of Belgium, France, Portugal, Spain, the Netherlands, or the UK, and 0 otherwise. Panel A reports statistics for the full sample, Panel B restricts to former colonies, and Panel C restricts to non-colonies. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and Sacks et al. (2010).

Appendix Table A.2: School-Harvest Overlap and Distance from the Inherited Vacation

	Dependent variable: School-harvest overlap O_i				
	(1)	(2)	(3)	(4)	(5)
Harvest distance to inherited vacation, months	0.027*** (0.009)	0.032*** (0.011)	0.034*** (0.010)	0.012 (0.015)	
Distance x similarity			0.105** (0.045)		
Calendar similarity, normalized			-0.300** (0.136)		
Distance x retained				0.041** (0.020)	
Calendar similarity above median				-0.105* (0.055)	
Harvest distance to colonizer harvest, months					0.030*** (0.009)
Slope, similarity above median s.e.				0.053 0.013	
Colonizer fixed effects	No	Yes	Yes	Yes	Yes
Observations	77	77	77	77	68

Notes. The sample is former colonies with nonmissing overlap and harvest data. Distance is the circular gap in months between the colony's main crop harvest midpoint and the midpoint of the colonizer's longest vacation. Calendar similarity is the share of the colonizer's vacation days on which the colony is also on vacation, and retained indicates similarity above the median among former colonies. Column 5 uses the colonizer's own harvest midpoint as the reference, available for France, Spain, and the United Kingdom only. Robust standard errors in parentheses. ***, **, * denote significance at the 1, 5, 10 percent level. Source: own elaboration based on data described in Appendix B.1.

Appendix Table A.3: Calendar Overlap and School Dropout Rates by Colonies and Non-Colonies

	Dependent variable: Dropout rate				
	(1)	(2)	(3)	(4)	(5)
Panel A: Main effects					
School-Harvest overlap	0.232*** (0.082)	0.167* (0.093)	0.209** (0.100)	0.207** (0.103)	0.209* (0.106)
Observations	120	120	120	120	120
Panel B: Colonies					
School-Harvest overlap	0.096 (0.116)	0.108 (0.115)	0.180 (0.121)	0.163 (0.131)	0.184 (0.133)
Observations	81	81	81	81	81
Panel C: Non-colonies					
School-Harvest overlap	0.065 (0.120)	0.031 (0.096)	0.026 (0.098)	-0.058 (0.070)	-0.052 (0.078)
Observations	39	39	39	39	39
Geographic controls	No	Yes	Yes	Yes	Yes
Macroeconomic controls	No	No	Yes	Yes	Yes
Demographic controls	No	No	No	Yes	Yes
Trade controls	No	No	No	No	Yes

Notes. The dependent variable is the out-of-school share, the share of primary-school-age children not enrolled in primary or secondary education. For each country, we use the latest available value during the 2010-2019 period. Geographic controls: country latitude. Macroeconomic controls: GDP, consumption share of GDP, and agricultural value added. Demographic controls: population. Trade controls: exports and the total trade volume between each country and the set of colonizers, comprising Belgium, France, Portugal, Spain, the Netherlands, and the UK. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and [Sacks et al. \(2010\)](#).

Appendix Table A.4: Descriptive Statistics

	Before		After	
	Affected	Unaffected	Affected	Unaffected
Panel A: Schooling outcomes				
No. students enrolled	78.66	77.57	70.68	75.42
Dropout rate	0.068	0.078	0.034	0.043
Grade promotion rate	0.844	0.819	0.874	0.847
Schools	1,668	16,885	1,668	16,885
Panel B: School-Harvest overlap (share of year, 2010)				
Overlap under Calendar A	0.196			
Overlap under Calendar B	0.194			
Change in overlap	0.002			
Change in overlap (crop level)	0.017			
Share with positive change	0.604			
Share with no change	0.092			
Share with negative change	0.305			
Panel C: Sample composition (2010)				
Share rural (all schools)	0.565	0.599		
Share public (all schools)	0.843	0.888		
Share within 40 km of boundary (bordering depts.)	0.769	0.493		
Share within 50 km of boundary (bordering depts.)	0.869	0.605		
Share within 60 km of boundary (bordering depts.)	0.940	0.712		
Panel D: Baseline municipality characteristics (2010)				
GDP per capita (thousands of pesos)	6,734	7,189		
Terrorist attacks	5.44	2.25		
Coca cultivation (ha)	116.92	99.13		
Municipal public expenditure	59,950	34,199		
Panel E: Teachers				
No. teachers	3.81	3.33	3.68	3.55
Share male teachers	0.322	0.307	0.336	0.327
Share teachers with postsecondary	0.721	0.648	0.827	0.751
Panel F: Crop harvesting (veredas)				
Harvested area (ha)	158.3	110.2		
No. of products	9.8	11.1		
Stimulants (ha)	28.2	24.6		
Wetland rice (ha)	26.0	13.0		
Maize (ha)	25.8	16.6		
Veredas	2,400	29,905		

Notes. Affected departments are Nariño and Valle del Cauca. Unaffected departments are the rest of the country. Before and After refer to the periods 2006-2010 and 2011-2018. Panels A and E report the average of each variable across schools in the estimation sample of rural public schools. Panel B reports calendar overlap measures as shares of the calendar year, computed for the baseline year 2010. The school-level change weights crops by each school's surrounding *vereda* crop composition, while the crop-level change is the unweighted mean across crop-by-department cells. Panel C reports the share of rural and public schools among all schools in the census, and the share of schools located within each distance of the boundary of the affected departments. Panel D reports baseline municipality characteristics in 2010, averaged across schools. Panel F reports the average harvested area and the average number of distinct crops across all *veredas* in affected and unaffected departments. Source: Own elaboration based on data from DANE, CEDE, and the Global Agro-Ecological Zones (GAEZ) v4, FAO.

Appendix Table A.5: Effect of Calendar Overlap on Schooling Outcomes: Dose-Response Estimates

	Dropout rate		Grade promotion		Enrollment (logs)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: All students						
School-Harvest overlap ($\Delta O_s \times \text{Post}$)	0.711*** (0.094)	0.740*** (0.096)	-0.654*** (0.144)	-0.691*** (0.143)	-2.169*** (0.609)	-1.984*** (0.592)
Observations	240,448	240,448	240,448	240,448	240,448	240,448
Panel B: Primary students						
School-Harvest overlap ($\Delta O_s \times \text{Post}$)	0.802*** (0.100)	0.829*** (0.102)	-0.699*** (0.153)	-0.742*** (0.152)	-2.364*** (0.592)	-2.024*** (0.578)
Observations	232,999	232,999	232,999	232,999	232,999	232,999
Panel C: Secondary students						
School-Harvest overlap ($\Delta O_s \times \text{Post}$)	-0.152 (0.190)	-0.079 (0.181)	-0.243 (0.337)	-0.319 (0.324)	1.177 (1.189)	0.582 (1.158)
Observations	30,979	30,979	30,979	30,979	30,979	30,979
Baseline controls $\times$ year FE	No	Yes	No	Yes	No	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the static dose-response estimate of Equation 5 on the all-country sample of rural public schools. Odd columns include school and year fixed effects only, even columns add interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure). Within each column pair, both specifications are estimated on the same sample of schools with nonmissing baseline controls. Panels B and C estimate the specification on the outcomes of primary and secondary students, each on its own balanced panel. Standard errors clustered at the school level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Own elaboration based on data from DANE.

Appendix Table A.6: Effect of School–Harvest Overlap on Grade Promotion Conditional on Completing the Year

	Grade promotion	Conditional promotion	
	(1)	(2)	(3)
School-Harvest overlap ($\Delta O_s \times \text{Post}$)	-0.707*** (0.142)	0.111 (0.082)	0.103 (0.081)
Pre-reform mean, treated	0.844	0.945	0.945
Observations	239,928	239,928	239,928
Baseline controls $\times$ year FE	Yes	Yes	Yes
Treated $\times$ year FE	No	No	Yes
School fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes

Notes. The table reports the static dose-response estimate of Equation 5. Column 1 uses the grade promotion rate, measured as the share of enrolled students who are promoted. Columns 2 and 3 use the promotion rate conditional on completing the school year, defined as the number of promoted students divided by the sum of promoted and failed students. All columns are estimated on the same balanced panel of rural public schools with nonmissing conditional promotion. Column 3 adds treated-by-year fixed effects. Because conditional promotion is measured among students who remained in school, estimates in columns 2 and 3 may reflect changes in the composition of students who stay. Pre-reform mean is the outcome mean in affected departments over 2006 to 2010. Standard errors clustered at the school level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Own elaboration based on data from DANE.

Appendix Table A.7: Effect of the Reform on Children’s Agricultural Work by Age

	Children’s agricultural work		
	(1)	(2)	(3)
Panel A: By age			
	Ages 10 to 15	Ages 10 to 12	Ages 13 to 15
Treated $\times$ Post	0.039*** (0.012)	0.030*** (0.011)	0.049** (0.020)
Pre-reform mean, treated	0.070	0.036	0.104
Effect relative to mean	0.56	0.83	0.47
Observations	19,746	9,750	9,996
Panel B: By school enrollment, ages 10 to 15			
		Enrolled	Not enrolled
Treated $\times$ Post		0.038*** (0.011)	0.089* (0.048)
Pre-reform mean, treated		0.044	0.263
Effect relative to mean		0.87	0.34
Observations		18,055	1,689
Baseline controls $\times$ year FE	Yes	Yes	Yes
Municipality fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Age and sex fixed effects	Yes	Yes	Yes

Notes. The table reports the coefficient on the interaction between an indicator for affected departments (Nariño and Valle del Cauca) and a post-reform indicator, from a static version of Equation 7, estimated separately by age group. The outcome is an indicator equal to one if the child was employed in the reference week in an agricultural or forestry activity, defined by two-digit CIIU branch. Control municipalities are in Risaralda and Sucre. All specifications absorb municipality, year, age, and sex fixed effects and interactions between year fixed effects and baseline municipality characteristics (per capita GDP, terrorist attacks, homicide rate, rural population, and agricultural GDP). Regressions are unweighted. Pre-reform mean is the outcome mean in affected departments over 2007 to 2010. Standard errors clustered at the municipality level, 93 clusters, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Own elaboration based on data from DANE.

B. Worldwide School and Agricultural Calendars

B.1. Data Construction

In this appendix, we describe the construction of the school and agricultural calendars for each country in our sample, as well as the measure of overlap between them.

School Calendar. We hand-collected academic calendar information from the official websites of ministries of education or equivalent government bodies in each country, using the most recently available calendar at the time of data collection. For each country, we recovered the start and end dates of the academic year, as well as the dates of all main school breaks. We then constructed a day-level indicator variable equal to one if school is in session on a given day of the year and zero otherwise.

For countries with multiple regional calendars, we selected the calendar of the most populous region or capital. Specifically, we used England for the United Kingdom, Berlin for Germany, the Central Region for the Netherlands, the French Community for Belgium, Washington D.C. for the United States, and Ontario for Canada. For Nigeria, India, and Indonesia, however, we deliberately deviated from this convention. Rather than selecting the capital region, we selected the calendar of the region that is the primary producer of the country's main crop as identified in the harvesting data: Kano state for Nigeria (sorghum), West Bengal for India (rice), and Central Java for Indonesia (rice). This choice ensures that the school calendar used in the overlap calculation reflects the institutional environment actually faced by farming households in the main crop-producing areas.

Harvesting Calendar. To construct harvesting calendars, we rely on the crop calendar dataset of [Sacks et al. \(2010\)](#), which reports the start and end dates of planting and harvesting seasons for nineteen major crops in each country. We define each country's main crop as the one with the largest national harvested area, using the harvested-area information distributed with the crop calendar dataset. For crops with multiple growing seasons, we use the main season as designated in the dataset. For a small number of countries not covered by these sources, we complement them with the FAO GIEWS country crop calendars.

Overlap Measure. Based on the school and harvesting calendars described above, we compute for each country the share of the year in which both school is in session and the main crop harvest is underway. Formally, this is the number of days on which both activities overlap, divided by 365. This measure captures the extent to which the institutional school calendar conflicts with the peak labor demand period in agriculture.

Educational Laws. To measure the number of years elapsed between the institutionalization of each colonizer's education system and the independence of each colony, we identify for each colonizer the foundational law that established public primary schooling as a state responsibility. These laws share a common logic: they replaced church-dominated or locally fragmented schooling with a centralized, state-regulated system of primary education, thereby creating the centralized institutional framework through which academic calendars were subsequently standardized and transmitted to colonial territories. We use the dates of these laws as markers of when each colonizer's centralized education system, of which the calendar became a component, was institutionalized.

For France, we use the Loi Guizot of 28 June 1833 (Loi sur l'instruction primaire), which was

the first law to create a durable national primary school system, requiring every municipality to maintain at least one school, establishing a mandatory curriculum, and creating a national teacher certification system and a network of teacher-training colleges. For Belgium, we use the *Loi Nothomb* of 23 September 1842 (*Loi organique de l'instruction primaire*), which was the founding act of Belgium's education system as an independent state, obliging every municipality to maintain at least one primary school and embedding religious instruction within the academic calendar. For the Netherlands, we use the *Schoolwet van 1806* (*Wet voor het Lager Schoolwezen en de Onderwijzers*), which, despite being preceded by two short-lived school acts in 1801 and 1803, was the first durable organic law of Dutch primary education, remaining in force for over fifty years and establishing classroom-based instruction, teacher examinations, and national school inspections as the pillars of a secular public school system. For Spain, we use the *Ley de Instrucción Primaria* of 21 July 1838 (*Ley Someruelos*) together with its *Reglamento* of 26 November 1838, which collectively established the first functioning liberal primary school system in Spain, requiring municipalities to maintain public schools, introducing teacher certification, and creating provincial supervision of education.¹ For Portugal, we use the *Carta de Lei* de 6 de Novembro de 1772, enacted under the Marquis of Pombal, which abolished Jesuit-controlled schooling, created a state-funded network of royal primary schools across Portugal and its empire, and introduced the *subsídio literário*—a dedicated national tax to fund teachers' salaries—marking the first systematic intervention of the Portuguese state in public primary education.² For the United Kingdom, we use the *Elementary Education Act* of 1870 (the *Forster Act*), which was the first law to establish a national framework for public primary schooling in England and Wales, filling the gaps left by the existing patchwork of voluntary and church schools by creating elected local school boards empowered to build and maintain publicly funded schools, levy local taxes for education, and—for the first time—work toward universal provision of elementary schooling across the country.

Independence Years. For each country in our sample, we use the year of independence from its corresponding colonizer. In most cases, this corresponds to the year in which the country formally declared or was granted independence. For a small number of cases, the year we use requires clarification. For Bangladesh, we use 1947—the year of the partition of British India and the creation of Pakistan, of which Bangladesh (then East Pakistan) was a part—rather than 1971, the year of Bangladesh's independence from Pakistan, since our interest is in the duration of exposure to the British institutional framework. For the Philippines, we use 1898—the year Spain ceded the territory following the Spanish-American War—rather than 1946, the year of formal independence from the United States, since our colonizer of interest is Spain and 1898 marks the effective end of Spanish institutional influence over the country. For the Dominican Republic, we

¹The institutional framework established by the *Ley Someruelos* was later consolidated and extended by the *Ley de Instrucción Pública* of 9 September 1857 (*Ley Moyano*), which introduced compulsory and free elementary schooling and integrated primary, secondary, and higher education into a single national system. While the *Ley Moyano* is often cited as the more comprehensive foundational law of Spanish education, we use the 1838 law as our reference date as it marks the first effective state intervention in primary schooling.

²Subsequent nineteenth-century reforms, notably those of 1822, 1835, and 1844, sought to reorganize and expand primary schooling in Portugal under liberal constitutional governments, but were largely left incomplete due to the country's severe economic difficulties and political instability during this period. We nonetheless retain the 1772 Pombaline reform as our reference date, as it constitutes the original act by which the Portuguese state assumed direct responsibility for public education, displacing the Jesuit institutional model that had governed schooling in Portugal and its overseas territories since the sixteenth century.

use 1865—the year of its final independence from Spain following the War of Restoration—rather than 1844, the year of its independence from Haiti, since our colonizer of interest is Spain. In all three cases, the years we use correspond to the moment at which the country ceased to be under the direct institutional influence of the colonizer whose education system we are studying, which is the relevant date for our analysis.

B.2. Agricultural and Institutional Timing across Latitudes

This subsection shows that harvest timing tracks geography, but school calendars, captured by vacation timing, do not. Appendix Figure A.1 shows that school calendars vary with latitude and colonizer. This figure maps monthly school-day distributions across the year (beginning in April and ending in March) for a global sample of countries sorted vertically from north to south. Countries are categorized by institutional origin to isolate the influence of colonial history from geographical location. It shows that, absent colonial influence, academic calendars shift predictably across latitudes—placing major breaks in July–August in the Northern Hemisphere and December–February in the Southern Hemisphere. However, we notice institutional inheritance may create systematic calendar misalignment: former British, French, Belgian, and Dutch colonies in tropical and southern latitudes predominantly retain European academic schedules despite local seasonal cycles.

Agricultural timing responds to geography. Panel a of Appendix Figure B.1 plots each country’s main-crop harvest midpoint against latitude. Harvest timing shifts systematically with distance from the equator and, most importantly, flips by roughly six months across it. Countries in the northern tropics concentrate their harvests toward the end of the calendar year, while countries in the mirror bands south of the equator harvest in the first months of the year, reflecting the reversal of solar seasonality between hemispheres. The dashed segments mark the average harvest midpoint in each hemisphere and make the six-month shift explicit.

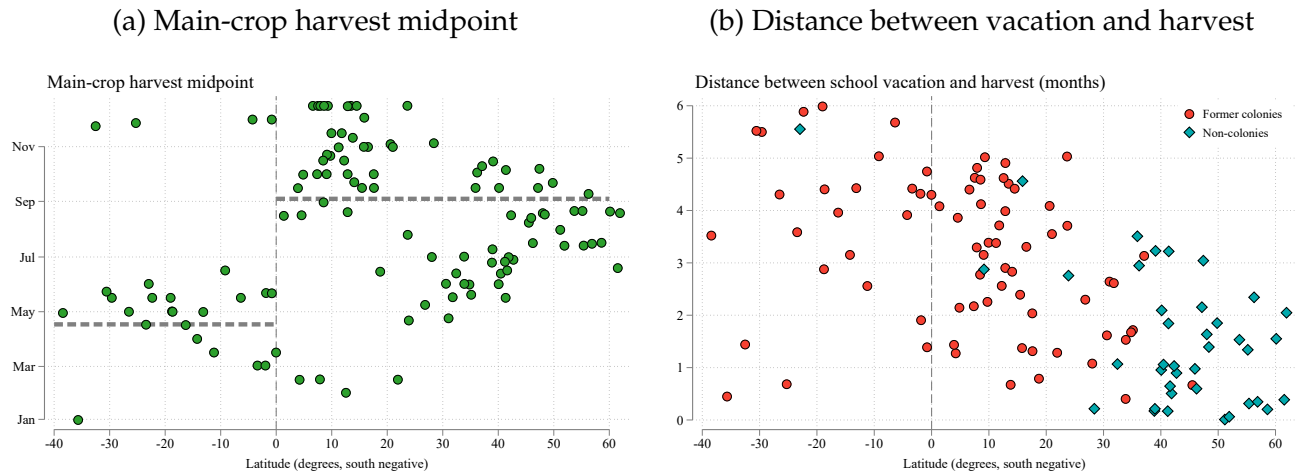
Institutional timing does not track geography in the same way. Panel b plots, for each country, the distance in months between the midpoint of the long school vacation and the main-crop harvest midpoint. In countries without a colonial past, which designed their own calendars, the long vacation sits close to the harvest, consistent with the historical origins of school calendars described in Section 2. In former colonies the vacation drifts away from the harvest, and the misalignment is largest south of the equator, where the average distance exceeds four months out of a maximum possible six. Notably, this is true even though most southern colonies have moved their long vacation to the austral summer. Because tropical harvests follow rainfall rather than solar seasons, adapting the inherited template to the local summer does not align it with the local harvest. Institutional adjustment followed the logic of the template, placing vacations in summer, rather than the agricultural conditions the original calendars were designed around.

Appendix Table B.1 summarizes the contrast in ten-degree latitude bands. In every band, the average vacation-to-harvest distance is larger in the Southern Hemisphere than in its northern mirror, and it is smallest in the temperate north, where school calendars originated. Together, the two panels illustrate the mechanism underlying our identification strategy. Harvest timing is predictable from geography, which is the premise of the instrument in Section 3, while the position of school vacations relative to harvests carries the imprint of institutional history.

Appendix Table B.1: Harvest and Vacation Timing by Latitude Band

Latitude band	Northern Hemisphere				Southern Hemisphere			
	N	Harvest	Vacation	Gap	N	Harvest	Vacation	Gap
0–10	18	Nov	Sep	3.3	9	Apr	Aug	4.0
10–20	18	Nov	Aug	3.0	7	May	Dec	3.9
20–30	9	Nov	Jul	2.7	6	May	Dec	4.3
30+	43	Aug	Jul	1.4	4	Mar	Jan	2.7

Notes: The table reports, for ten-degree bands of latitude in each hemisphere, the number of sample countries, the typical month in which the main crop is harvested, the typical month of the midpoint of the school vacation period, and the average distance in months between the school vacation and the harvest. The distance ranges from zero, when the vacation falls on the harvest, to six months, when it falls in the opposite part of the year. All midpoints and distances are computed treating the calendar year as a circle, so that seasons spanning December and January are measured correctly. Source: Own elaboration based on data described in Appendix B.1.

Appendix Figure B.1: Agricultural and Institutional Timing across Latitudes

Notes: Panel a plots the midpoint of each country's main-crop harvest season against latitude, with southern latitudes negative. The horizontal dashed segments mark the average harvest midpoint in each hemisphere. Panel b plots the distance in months between the midpoint of each country's school vacation period and its harvest midpoint, which ranges from zero, when the vacation falls on the harvest, to six months, when it falls in the opposite part of the year. All midpoints and distances are computed treating the calendar year as a circle, so that seasons spanning December and January are measured correctly. The vertical dashed line marks the equator. Source: Own elaboration based on data described in Appendix B.1.

B.3. Robustness to the Instrumental Variable Strategy

Table B.2 examines whether the cross-country estimates reflect institutional or geographic characteristics correlated with the instrument. Starting from the specification in column 4 of Table 3, each column adds one block of pre-determined controls. Column 2 includes colonizer fixed effects, so that identification comes only from comparisons among colonies of the same colonial power, which inherited the same school calendar but face different harvest timing. Column 3 adds World Bank region fixed effects, column 4 adds fixed effects for the latitude zones used to construct the instrument, and column 5 controls for religious fractionalization (Alesina et al., 2003). Among former colonies, the coefficients range from 0.47 to 0.74, and in the full sample from 0.51 to 0.86. The Anderson–Rubin confidence sets exclude zero in every specification of both panels.

Appendix Table B.2: Robustness of the Cross-Country Estimates to Pre-Determined Institutional and Geographic Controls

	Dependent variable: Dropout rate				
	(1)	(2)	(3)	(4)	(5)
Panel A: Former colonies					
School-Harvest overlap	0.630** (0.305)	0.562* (0.300)	0.470 (0.338)	0.736* (0.395)	0.686** (0.315)
Effective first-stage F	11.18	9.00	6.82	8.40	9.70
Partial first-stage R^2	0.256	0.225	0.181	0.193	0.227
AR 95% confidence set	[0.18, 1.79]	[0.14, 1.84]	[0.01, 2.28]	[0.18, 2.54]	[0.23, 1.98]
Observations	81	81	81	81	80
Panel B: Full sample					
School-Harvest overlap	0.785*** (0.251)	0.511* (0.264)	0.571** (0.253)	0.765** (0.384)	0.860*** (0.258)
Effective first-stage F	17.49	10.37	11.91	9.04	15.40
Partial first-stage R^2	0.238	0.207	0.182	0.130	0.224
AR 95% confidence set	[0.38, 1.55]	[0.12, 1.55]	[0.20, 1.50]	[0.15, 2.28]	[0.46, 1.70]
Observations	120	120	120	120	119
Colonizer fixed effects	No	Yes	No	No	No
Region fixed effects	No	No	Yes	No	No
Latitude-zone fixed effects	No	No	No	Yes	No
Religious fractionalization	No	No	No	No	Yes

Notes. The dependent variable is the out-of-school share. All columns are estimated by two-stage least squares with Z_i as the instrument and include the controls of column 4 of Table 3. Each column adds one block of controls to that specification. Colonizer fixed effects are indicators for the six colonial powers, plus the never-colonized group in Panel B. Region fixed effects are indicators for the seven World Bank regions. In Panel A, Europe and Central Asia contains a single former colony, so its region indicator absorbs that observation. Latitude-zone fixed effects are indicators for the latitude zones used to construct the instrument. Religious fractionalization is from Alesina et al. (2003) and is not available for the Cayman Islands. The effective first-stage F is the statistic of Montiel Olea and Pflueger (2013), whose critical value for a worst-case bias of 10 percent is 23.1 with a single instrument. The partial first-stage R^2 is the share of the variation in school-harvest overlap explained by the instrument after partialling out the controls. The AR 95% confidence set contains the values of the coefficient not rejected at the 5 percent level by a heteroskedasticity-robust Anderson and Rubin (1949) test. ***, **, * denote significance at the 1, 5, 10 percent significance level. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, Sacks et al. (2010), and Alesina et al. (2003).

We also assess the robustness of our results in Section 3 by introducing a second instrument that exploits the interaction between the school calendar of the colonizing country, the agricultural calendar predicted by their latitude, and the degree of exposition to colonial education institutions.

Let $c(i)$ denote the country that colonized country i and let $\mathbb{L}_i$ denote the set of countries located at latitudes similar to that of country i . We construct the following instrument:

$$Z_i^C = \left(\frac{1}{|\mathbb{D}|} \sum_{d \in \mathbb{D}} S_{c(i),d} \hat{p}_{id} \right) \times \frac{Y_i^{indep}}{Y_{c(i)}^{educ}},$$

where $S_{c(i),d}$ is an indicator equal to one if schools are in session in the colonizing country $c(i)$ on day d , $\hat{p}_{id} \in [0, 1]$ is the leave-one-out predicted probability that a country with the agricultural conditions of country i is harvesting on day d , as defined in Section 3, and the final term is the ratio of the year in which country i gained independence to the year in which the colonizer institutionalized its school calendar. The ratio exceeds one for colonies that remained under colonial rule after the colonizer’s calendar was institutionalized, and increases with the length of that post-institutionalization exposure, so the instrument places relatively more weight on countries whose school systems developed while the colonizer’s calendar template was already in place. Using this instrument, we estimate Equation 1.

This alternative instrument combines two sources of variation in a manner similar to a shift-share design (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2021). Because the instrument is the product of the two components, its validity rests on the excludability of the interaction itself, as we discuss below. The plausibility of that requirement, in turn, draws on the properties of each component.

The *share* component corresponds to the same used in Section 3, and quantifies the overlap that would arise if former colonies adopted the school calendar of their colonizer while facing agricultural conditions determined by their latitude. This term isolates variation in school calendars inherited from the colonizer while accounting for local agricultural cycles. Its exogeneity is plausible because colonial powers originally designed their domestic school calendars to reflect their own climatic and agricultural conditions rather than those of the territories they later administered. Moreover, by using agricultural calendars derived from countries located at similar latitudes, the instrument captures agricultural conditions predicted by geography rather than country-specific institutional choices.

The *shift* component captures the extent to which colonial institutions influenced the determination of school calendars in country i . Specifically, it reweights the inherited calendar variation according to the timing of independence relative to the educational reform in the colonizing country. Countries that gained independence later were more likely to inherit the colonizer’s calendar, while those that became independent earlier had greater scope to adopt calendars adapted to local conditions. This timing is plausibly exogenous to current educational outcomes because the date of colonial independence relative to the timing of educational reforms in the colonizing country was largely determined by geopolitical and historical forces unrelated to contemporary school dropout.

Taken together, this second instrument captures the overlap that country i would have experienced if it had adopted the colonizer’s calendar while facing agricultural conditions determined by its latitude, with relatively more weight placed on countries that remained under colonial rule after the colonizer institutionalized its calendar. The validity of this instrument requires that the interaction itself be excludable. Conditional on the controls, the product of the colonizer-calendar overlap and the length of colonial exposure must affect dropout only through the inherited

school calendar, which is a different, and not weaker, requirement than the exogeneity of either component alone. Its plausibility rests on the same logic as the baseline instrument, with exposure duration entering as an intensity weight. Colonizers institutionalized their calendars for their own conditions, and a longer period of colonial rule after that institutionalization gave the transplanted calendar more time to take root. Because the specifications control for latitude directly, the instrument's identifying variation does not reduce to geographic location. We therefore read the estimates in Table B.3 as a robustness check under an alternative exposure weighting rather than as an independently identified estimate.

Table B.3 presents the results of the estimations using this version of the instrument. In the full sample, the coefficients range from 0.81 to 0.84 in columns 1 to 4, close to the baseline estimates, and fall to 0.62 when we control for colonial status in column 5. Among former colonies, the coefficients remain positive and statistically significant, ranging from 0.56 to 0.68, somewhat below the full-sample estimates but consistent with them in sign and significance. The Anderson–Rubin confidence sets exclude zero in every specification of both panels. Our results are therefore robust to this alternative exposure weighting of the instrument.

Appendix Table B.3: Effect of School–Harvest Overlap on the Out-of-School Share: Exposure-Weighted Instrument

	Dependent variable: Dropout rate				
	(1)	(2)	(3)	(4)	(5)
Panel A: Full sample					
School-Harvest overlap	0.837*** (0.292)	0.821*** (0.259)	0.811*** (0.252)	0.824*** (0.259)	0.615** (0.290)
Effective first-stage F	18.35	17.59	17.92	16.79	11.71
Partial first-stage R^2	0.228	0.227	0.243	0.240	0.229
AR 95% confidence set	[0.40, 1.78]	[0.42, 1.63]	[0.42, 1.59]	[0.42, 1.64]	[0.19, 1.72]
Observations	120	120	120	120	120
Panel B: Former colonies					
School-Harvest overlap	0.558* (0.302)	0.649** (0.301)	0.646** (0.293)	0.684** (0.313)	
Effective first-stage F	11.67	11.34	11.76	10.50	
Partial first-stage R^2	0.257	0.250	0.274	0.260	
AR 95% confidence set	[0.12, 1.75]	[0.21, 1.83]	[0.22, 1.77]	[0.24, 1.96]	
Observations	81	81	81	81	
Geographic controls	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	No	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes	Yes
Trade controls	No	No	No	Yes	Yes
Colonial status control	No	No	No	No	Yes

Notes. The dependent variable is the out-of-school share, the share of primary-school-age children not enrolled in primary or secondary education. For each country, we use the latest available value during the 2010–2019 period. Geographic controls: country latitude. Macroeconomic controls: GDP, consumption share of GDP, and agricultural value added. Demographic controls: population. Trade controls: exports and the total trade volume between each country and the set of colonizers, comprising Belgium, France, Portugal, Spain, the Netherlands, and the UK. The colonial status control is an indicator for former colonial status, included in both stages. It applies to Panel A only, since it is constant within the sample of former colonies. The effective first-stage F is the statistic of [Montiel Olea and Pflueger \(2013\)](#), whose critical value for a worst-case bias of 10 percent is 23.1 with a single instrument. The partial first-stage R^2 is the share of the variation in school–harvest overlap explained by the instrument after partialling out the controls. The AR 95% confidence set contains the values of the coefficient not rejected at the 5 percent level by a heteroskedasticity-robust [Anderson and Rubin \(1949\)](#) test, and is valid regardless of instrument strength. ***, **, * denote significance at the 1, 5, 10 percent significance level. Source: own elaboration based on administrative records, World Bank Indicators, Penn World Table version 11.0, and [Sacks et al. \(2010\)](#).

C. Historical Evolution of the Academic Calendar in Colombia

The academic calendar in Colombia has undergone substantial transformations since the early nineteenth century. These changes reflect shifting priorities in educational administration, evolving degrees of centralization, regional heterogeneity, and growing recognition of climatic and institutional differences across the country. From an initially centralized system that tightly regulated vacations and examinations nationwide, Colombia progressively moved toward differentiated academic calendars across regions—later formalized as Calendar A and Calendar B—before adopting a decentralized framework that allows territorial entities to define calendars within national guidelines.

C.1. Colonial Period (Before 1820s)

During the colonial period, particularly before the eighteenth century, education in Spain and its colonies was primarily provided by the Church in a decentralized manner. Since the sixteenth century, instruction was largely delivered by Catholic religious orders—such as Jesuits, Franciscans, and Dominicans—who focused mainly on teaching Christian doctrine and basic literacy in Indigenous languages, while learning Spanish was often optional. Education was typically delivered in *escuelas de primeras letras*, institutions that provided basic instruction by introducing children to Christian principles and values, teaching them to read, familiarizing them with the four basic arithmetic operations, and developing writing skills through exemplars and guided practice (Anguita Osuna, 2019). Over time, however, strong pressure (especially from archbishops, the Council of the Indies, and Crown officials) emerged to promote instruction in Spanish, as Indigenous languages were increasingly viewed as obstacles to the diffusion of Christianity.

During the eighteenth century, the Spanish Crown introduced the Bourbon reforms to modernize state capacity in the colonies (Chiovelli et al., 2023). As part of these reforms, and influenced by Enlightenment ideas, Carlos III promoted educational reforms that reduced the Church's influence over schooling and expanded free primary instruction in Spain (Anguita Osuna, 2019), and colonial authorities pursued parallel efforts to broaden educational access in the Americas (Herrera, 2004; García, 2005). This process gained momentum after the suppression of the Society of Jesus, which entailed the expulsion of Jesuit members from Spain and its colonies in 1767. Nevertheless, education remained accessible only to a small elite, with broad segments of society excluded (García, 2005). Instructional institutions were scarce in the Kingdom of New Granada (which encompassed present-day Colombia), as evidenced by the presence of only one free *escuela de primeras letras* in Santa Fe, the capital of the kingdom and the site of present-day Bogotá (García, 2005). Importantly, school calendars during this period played little role in shaping educational access.

C.2. Early Republican Regulation and Centralization (1820s–1870)

The landscape of Colombian society in the early years of national consolidation highlighted the need for a new and more structured educational system. Independence leaders promoted the creation of civic educational institutions, such as *colegios* and public schools, framed within

republican ideals of citizenship and Enlightenment thought, although resources and administrative capacity were limited. Efforts were made to implement laws and establish institutions that expanded access and provided practical instruction, but their reach remained uneven, and religious actors continued to exert significant influence (Báez Osorio, 2006).

Historical overviews suggest that school calendars were not a major policy concern in the early republic (Báez Osorio, 2006). Educational provision was still being organized, and formal structures such as fixed academic calendars were not yet nationally institutionalized.

Nonetheless, the earliest national regulation potentially affecting the academic calendar dates to 1824, when Vice President Francisco de Paula Santander issued a decree establishing that school vacations should coincide with the recesses observed by courts and tribunals, as defined in the Law of October 12, 1821. Under Articles 70 and 71 of that law, judicial activity was suspended only during major religious feast days, Holy Week, and December. Consequently, school vacations were institutionally anchored to these same periods.

This framework was reinforced by the General Law of 1826, which explicitly defined the structure of the academic year. Article 112 stipulated that annual vacations would run from the conclusion of examinations in November until January 1, and that no additional recesses would be permitted beyond religious feast days and Easter. Contemporary correspondence illustrates that this reform curtailed earlier local practices. In December 1826, students from the *Colegio de Boyacá* protested the elimination of a traditional vacation period running from late July to mid-October, which was replaced by a single end-of-year recess in December following the enactment of Santander's legislation. The governor of Boyacá responded that the matter fell under central authority, and the reform was ultimately reaffirmed.

This episode indicates that, prior to the reforms promoted under Santander's administration, academic calendars in at least some institutions followed a timing that differed substantially from the post-1826 national standard, including a long mid-year interruption with instruction extending across the calendar year. Because the colonial record described above contains no formalized term schedule, whether this earlier arrangement derived from Spanish-period practice cannot be established from the available sources. What the episode documents is that, from 1826 onward, such local arrangements were explicitly displaced by a centrally imposed calendar more closely resembling the January-November structure.

The emphasis on standardization continued with the Organic Decree of Public Instruction of 1870, which formalized a system of two academic terms per year, each lasting five months. Schools were to open on January 10 and July 10 and close on May 31 and November 30, with vacations in June and December. This decree established a nationally uniform structure that would remain influential for decades.

C.3. Consolidation and Incremental Adjustments (1880s–1890s)

Late nineteenth-century sources indicate both continuity and incremental adjustments within this centralized framework. Institutional correspondence from 1888 shows that intermediate examinations were held in late July, followed by short vacations, while final examinations took place in November. National decrees in 1892 and subsequent regulations consistently defined the academic year as running from February (or January) to November, with vacations concentrated in December–January.

However, some institutional and regional variation emerged. For example, the *Colegio Público de Barranquilla*, established in 1892, followed an academic year running from August to June. Regulations for normal schools and primary schools in 1893 retained the February–November or January–November structure but clarified enrollment periods, examination timing, and paid vacation months. These variations, however, remained exceptions within an otherwise standardized national system.

C.4. Departmental Differentiation in the Early Twentieth Century (1900s–1940s)

By the early twentieth century, departmental regulations increasingly reflected differentiated academic timing. Resolutions from 1906 illustrate this divergence clearly. In the Department of Cauca, which at the time still included the territory that in 1910 became the department of Valle del Cauca, secondary and professional institutions were required to operate on an academic year running from October 1 to July 20. By contrast, in the Department of Huila the school period was fixed from January 10 to November 30, with additional vacations during Holy Week and in late July.

A significant event occurred in 1913, when the Governor of Antioquia decreed that the academic year should begin in mid-September and end in June. The decree explicitly justified this structure by reference to pedagogical considerations, climatic conditions, and the convenience of students pursuing studies abroad.

Regional differentiation became increasingly pronounced in the southwest. Decree 1457 of 1939 required the Rural Normal School of Popayán to suspend classes at the end of July and resume them in October. Decree 3087 of 1945 further confirmed that institutions in the southwest operated under a calendar distinct from that followed in other regions. By this period, Colombia effectively maintained multiple regional academic calendars, though they had not yet been formally labeled.

C.5. Formal Recognition of Dual Calendars (1950s–early 1960s)

The regional division of academic calendars became explicit and institutionalized in the mid-twentieth century. Decree 0075 of 1951 clearly differentiated academic timing across regions: colleges in Valle del Cauca and Nariño were to implement the new curriculum beginning in October, while institutions in the rest of the country would do so beginning in February. This marked the first nationwide decree to formally recognize a dual calendar structure aligned with regional geography.

An attempt to unify the system followed with Decree 3109 of 1956, which mandated the unification of calendars beginning in September 1958. However, subsequent amendments—particularly Decree 1254 of 1957—introduced transitional adjustments, reflecting practical difficulties in implementing uniformity.

Rather than eliminating differentiation, reforms in the early 1960s codified it more precisely. Decree 45 of 1962 and Decree 1710 of 1963 established detailed academic periods, vacations, and examination schedules separately for the central-eastern and south-western regions, covering both secondary and primary education. These distinctions were further reinforced by Decree 2720 of 1963, which formally distinguished Calendar A (central-eastern region) and Calendar B (south-western region) and explicitly allowed schools to choose which calendar to follow.

C.6. Attempts at Unification and Reinstatement of the Dual System (Late 1960s–1980s)

A renewed effort to impose national uniformity came with Decree 1816 of 1967, which standardized instructional days and vacation lengths nationwide while allowing calendars to differ only in start dates. This reform was implemented gradually between 1968 and 1969.

However, the reform proved unsustainable. Decree 1902 of 1969 reversed course, citing reduced end-of-year vacations, administrative constraints in educational planning, the need for mid-year teacher training, climatic conditions in the southwest, and requests from school communities. The decree reinstated a clear territorial division: Calendar B for Nariño, Cauca, Valle del Cauca, and Putumayo, and Calendar A for the rest of the country.

This dual structure persisted through the 1970s and was reaffirmed in subsequent regulations. Decree 691 of 1972 established a provisional calendar following a nationwide suspension of activities, while Decree 174 of 1982 maintained Calendars A and B for both public and private institutions at the preschool and primary levels.

C.7. Decentralization and Contemporary Framework (2000s–Present)

A major institutional shift occurred with the decentralization reforms of the early 2000s. Law 715 of 2001 assigned departments responsibility for organizing education provision in non-certified municipalities, including defining the academic calendar, while granting certified municipalities autonomy over education provision. Decree 1850 of 2002 required certified territorial entities to issue annual academic calendars for all public institutions within their jurisdiction, explicitly allowing calendars to account for regional economic conditions and institutional traditions.

While this framework preserved the historical distinction between Calendars A and B, it also enabled greater coordination and gradual convergence in academic timing, particularly in certified and urban jurisdictions. As a result, contemporary academic calendars reflect both long-standing regional differentiation and increased institutional coordination within a decentralized governance structure.

D. Robustness of the School-Level Estimates

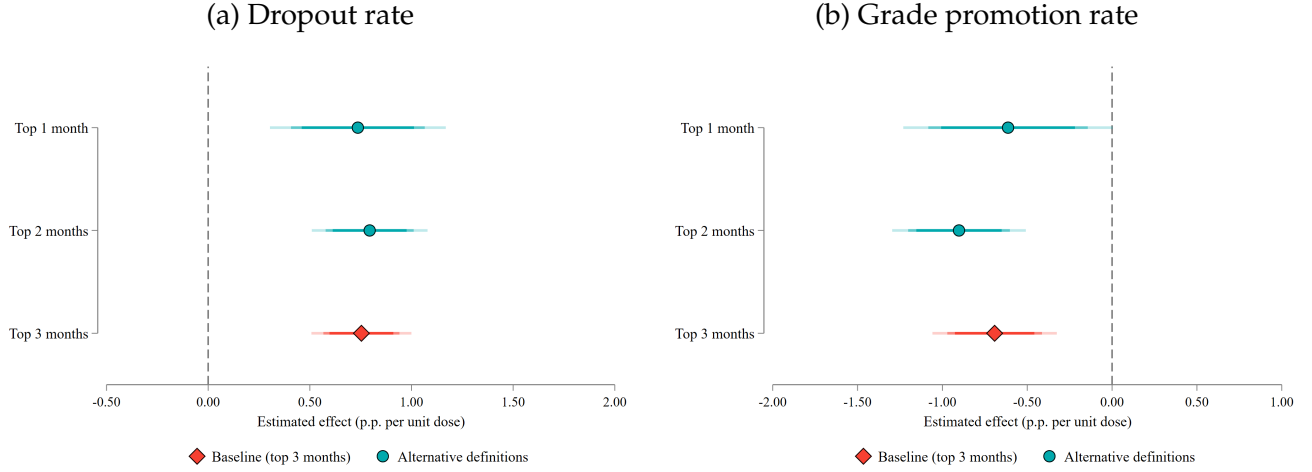
We provide additional evidence supporting our main results through several complementary exercises. We begin by showing that the effects on school progression are insensitive to alternative definitions of harvesting intensity. Then, we re-estimate the dose-response effects using alternative estimators that are robust to treatment-effect heterogeneity, restrict the control group to progressively more local comparisons, test the identifying assumption of parallel pre-trends across every specification, and summarize the full battery in specification-curve form. Finally, we present aggregate synthetic difference-in-differences estimates of the reform's intent-to-treat effects.

D.1. Alternative Definitions of Harvesting Intensity

Recall from Section 6.1 that the overlap measure for school s under calendar c is

$$O_{sc} = \sum_{j \in J} \omega_{sj} \left(\frac{1}{12} \sum_{m=1}^{12} \max \left\{ \mathbb{1}_{\{m \in \mathcal{I}_j\}} - V_{mdc}, 0 \right\} \right), \quad (\text{D.1})$$

where $\mathcal{I}_j$ corresponds to the three most intensive harvesting months, the top quartile of the annual distribution of harvesting for crop j . This threshold is arbitrary. We therefore re-estimate our main specification varying $\mathcal{I}_j$ to include only the two most intensive months and only the single most intensive month. Appendix Figure D.1 reports the results, with the baseline measure included as a reference. The effects retain their magnitude, sign, and remain statistically significant under all three definitions.

Appendix Figure D.1: Effects Varying the Definition of Harvesting Intensive Months ($\mathcal{I}_j$)

Notes: Each panel reports the static dose-response estimate of Equation 5 under alternative definitions of harvesting intensity, for the dropout rate (Panel a) and the grade promotion rate (Panel b). In each panel, the dose ΔO_s is constructed from the single most intensive harvesting month of the year (Top 1), the two most intensive months (Top 2), and the three most intensive months (Top 3), the latter corresponding to the baseline definition used throughout the paper and displayed as a diamond. All specifications are estimated on the balanced all-country panel of rural public schools between 2006 and 2018 and include school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics. Horizontal bars indicate 90%, 95%, and 99% confidence intervals based on standard errors clustered at the school level. Source: Own elaboration based on data from DANE.

D.2. Specification Curves

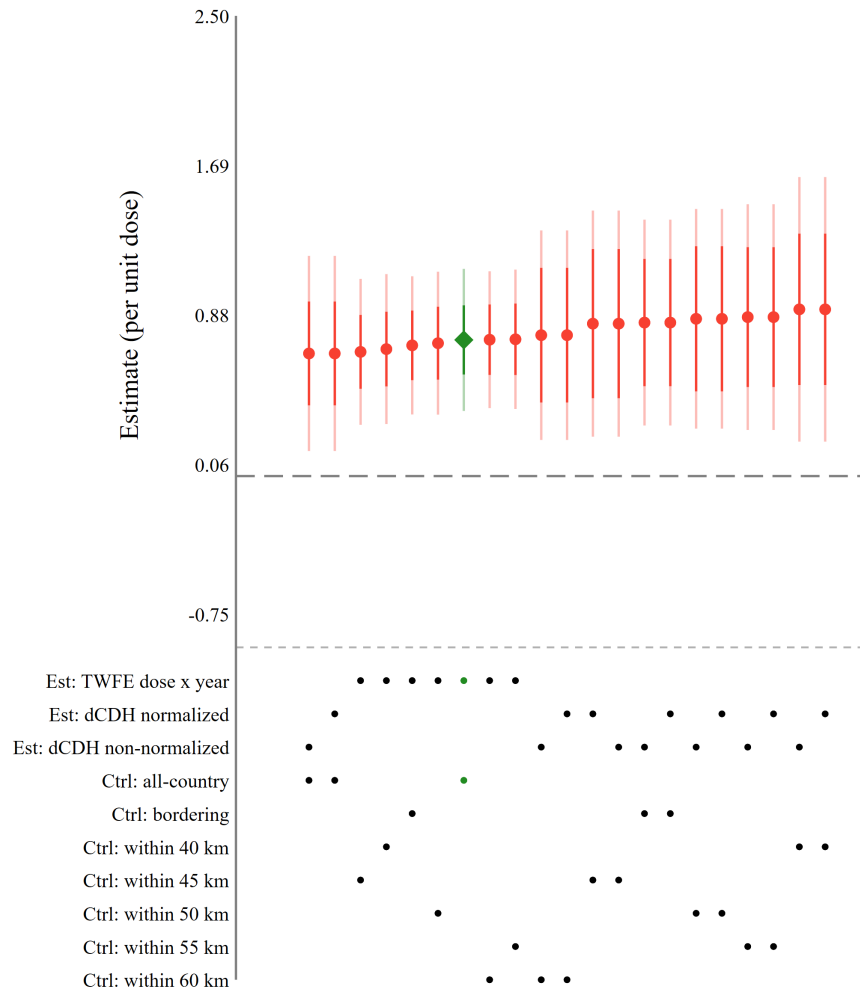
We estimate the static dose-response effect of the calendar reform for each of our outcomes (and, for completeness, also for school enrollment, measured in logs) under every combination of three estimators and seven control groups, for a total of 21 specifications per outcome.

The three estimators are: (i) the two-way fixed effects (TWFE) specification described in Section 6; (ii) the heterogeneity-robust estimator of [de Chaisemartin et al. \(2024\)](#) in its normalized version, which expresses the effect per unit of treatment dose; and (iii) the same estimator in its non-normalized version. All estimators include school and year fixed effects. Trend adjustment enters each estimator in its natural form. The TWFE specification interacts year fixed effects with baseline municipality characteristics (per capita GDP, terrorist attacks, coca cultivation, and municipal public expenditure), while the [de Chaisemartin et al. \(2024\)](#) estimator conditions on nonparametric cells defined by within-sample median splits of the same characteristics.

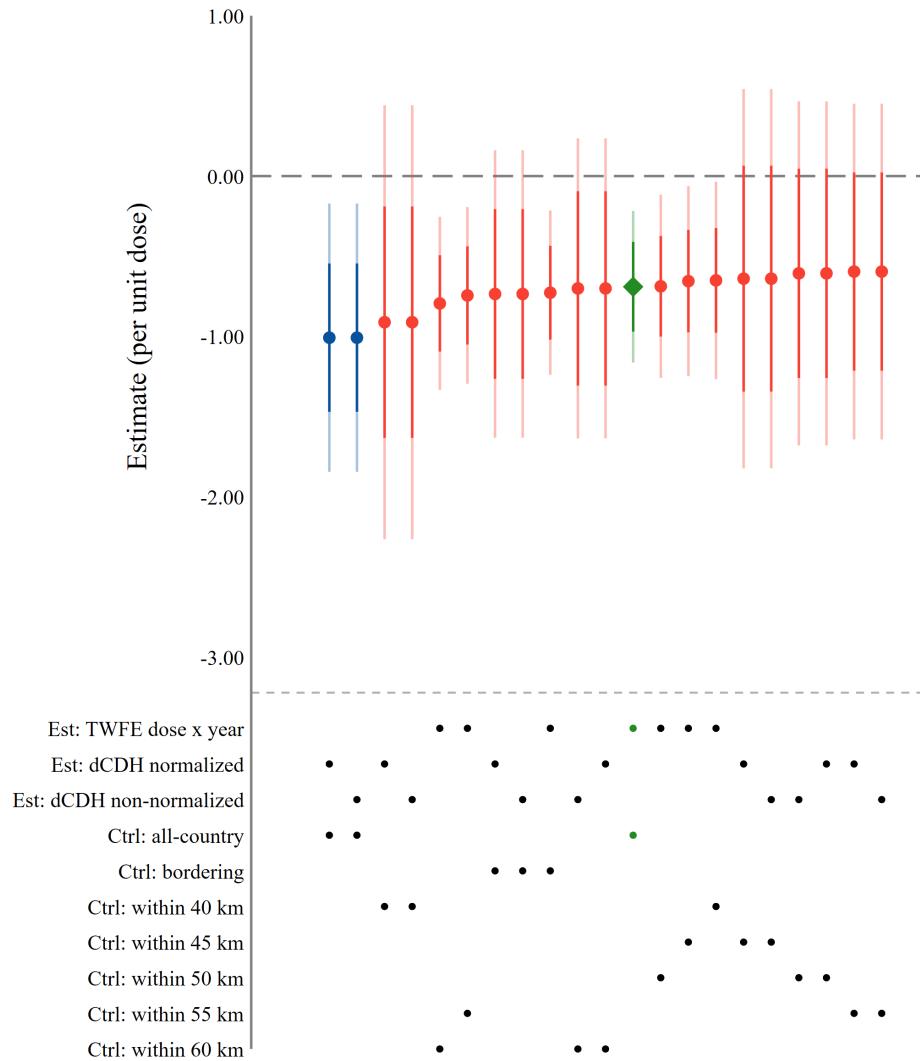
The seven control groups are: all rural public schools in the country; schools in the six departments bordering the treated ones; and schools located within 40, 45, 50, 55, and 60 kilometers of the boundary of the treated departments, based on georeferenced school locations. The distance-based groups narrow the comparison progressively toward the boundary, so that treated and control schools increasingly share local geographic and economic conditions. Each band applies to both sides of the boundary, restricting treated and control schools alike. The narrowest band, 40 kilometers, is close to the median distance to the boundary among control schools in the bordering departments (41 kilometers). It therefore retains the closer half of these schools and about three quarters of treated schools, whose median distance to the boundary is 22 kilometers.

Figures D.2, D.3, and D.4 present the resulting specification curves for dropout, grade promotion, and log enrollment. In each figure, the upper panel plots the 21 point estimates sorted by magnitude, with nested 95 percent confidence intervals clustered at the school level (solid) and at the municipality level (translucent). The lower panel indicates the estimator and control group corresponding to each specification. Each specification is colored according to the outcome of its joint pre-trend test: specifications in red do not reject the null hypothesis of no differential pre-trends at the 10 percent level, specifications in light blue reject at the 10 but not the 5 percent level, and specifications in dark blue reject at the 5 percent level.

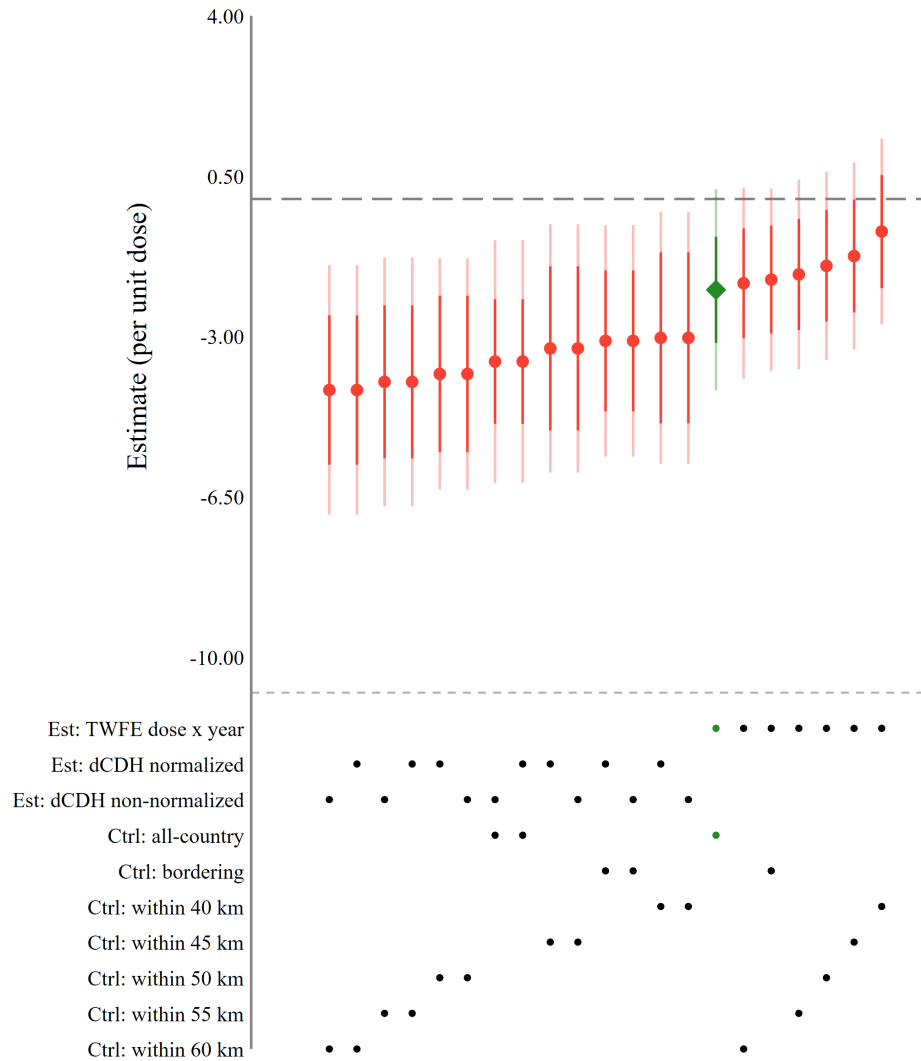
Three patterns emerge. First, the effect on dropout is positive and statistically significant at the 5 percent level in all 21 specifications, at both clustering levels, with estimates ranging from 0.59 to 0.90 percentage points per percentage point of overlap. Second, the effect on grade promotion is negative in all 21 specifications and statistically significant for most of them. Third, the effect on enrollment is negative in all specifications, ranging from 0.7 to 4.2 percent per percentage point of overlap, and statistically significant at conventional levels with school-level clustering in 20 of them. The only insignificant estimate corresponds to the narrowest control group (schools within 40 kilometers of the treated boundary), where the sample is smallest. No estimator or control group produces estimates that deviate from these patterns, indicating that the results are not an artifact of any particular modeling choice.

Appendix Figure D.2: Specification Curve: Dropout Rate

Notes: The upper panel plots the 21 static dose-response estimates for the dropout rate obtained by crossing three estimators with seven control groups, sorted by magnitude. Estimates are expressed per unit of the change in calendar overlap, ΔO_s . Solid vertical bars are 95% confidence intervals clustered at the school level; translucent bars are 95% confidence intervals clustered at the municipality level. The lower panel indicates the estimator and control group of each specification. Specifications in red do not reject the null hypothesis of no differential pre-trends at the 10 percent level; specifications in light blue reject at the 10 but not the 5 percent level; specifications in dark blue reject at the 5 percent level. Source: Own elaboration based on data from DANE.

Appendix Figure D.3: Specification Curve: Grade Promotion

Notes: The upper panel plots the 21 static dose-response estimates for the grade promotion rate. Estimates are expressed per unit of the change in calendar overlap, ΔO_s . Solid vertical bars are 95% confidence intervals clustered at the school level; translucent bars are 95% confidence intervals clustered at the municipality level. The lower panel indicates the estimator and control group of each specification. Specifications in red do not reject the null hypothesis of no differential pre-trends at the 10 percent level; specifications in light blue reject at the 10 but not the 5 percent level; specifications in dark blue reject at the 5 percent level. Source: Own elaboration based on data from DANE.

Appendix Figure D.4: Specification Curve: Enrollment (in logs)

Notes: The upper panel plots the 21 static dose-response estimates for log enrollment. Estimates are expressed per unit of the change in calendar overlap, ΔO_s . Solid vertical bars are 95% confidence intervals clustered at the school level; translucent bars are 95% confidence intervals clustered at the municipality level. The lower panel indicates the estimator and control group of each specification. Specifications in red do not reject the null hypothesis of no differential pre-trends at the 10 percent level; specifications in light blue reject at the 10 but not the 5 percent level; specifications in dark blue reject at the 5 percent level. Source: Own elaboration based on data from DANE.

D.3. Common Shocks and Within-Treated Identification

Because the dose ΔO_s is set to zero for all schools outside the affected departments, the dose-response coefficient combines variation in exposure among affected schools with the comparison between affected and unaffected schools. A post-reform shock common to all schools in Nariño and Valle del Cauca is therefore not mechanically excluded from the estimates. Table D.1 addresses this concern directly. It adds a full set of treated-by-year fixed effects to the baseline specification, absorbing any shock common to affected schools and identifying the coefficient exclusively from variation in exposure among them. The estimates for dropout and grade promotion are essentially unchanged, 0.75 and -0.75 per percentage point of overlap, indicating that the comparison with unaffected schools contributes little to the headline effects. The enrollment estimate falls by about half, consistent with enrollment absorbing scale and demographic components common to affected departments rather than reflecting the overlap channel alone.

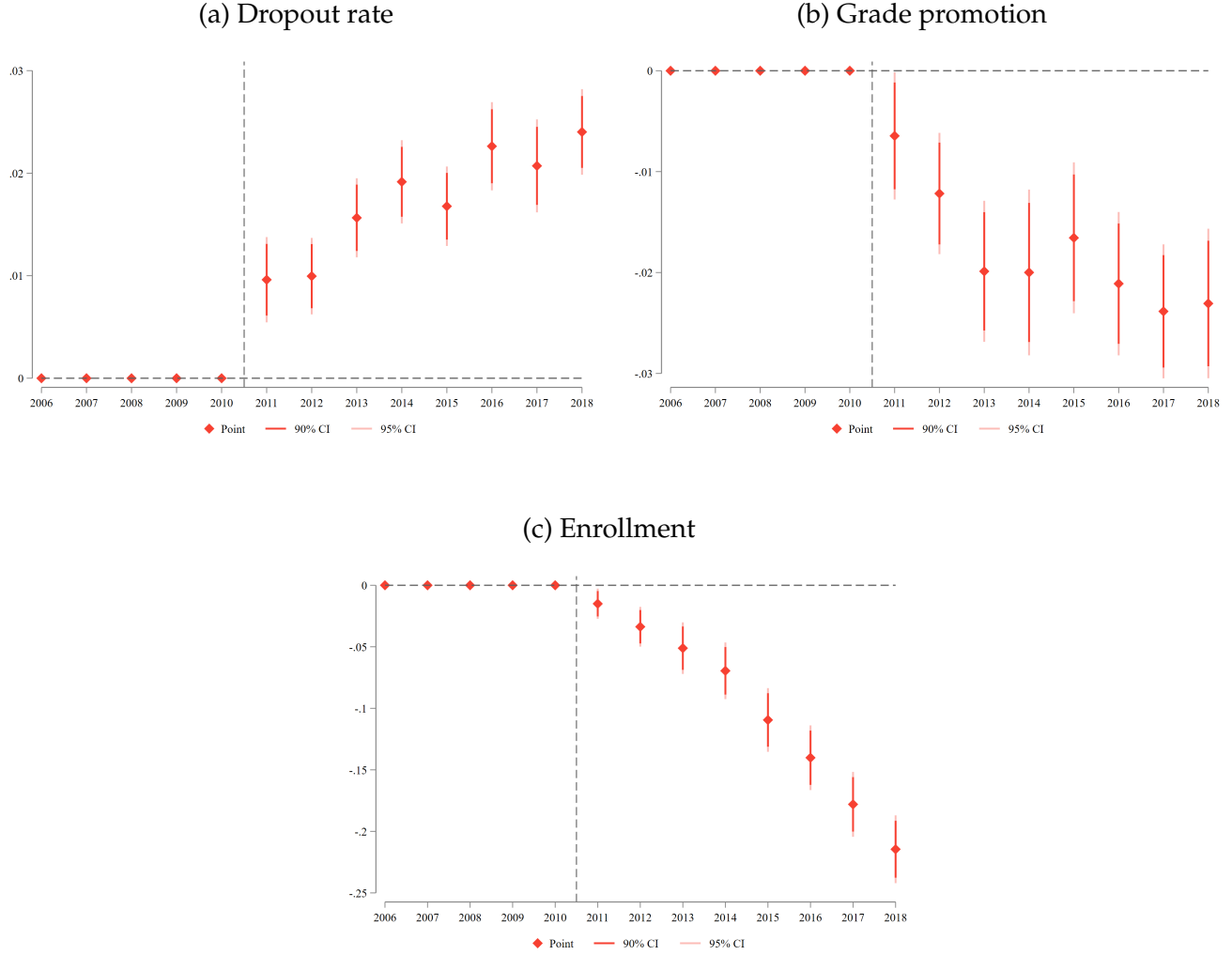
Appendix Table D.1: Identification from Within-Treated Variation

	Dropout rate (1)	Grade promotion (2)	Enrollment (logs) (3)
$\Delta O_s \times \text{Post}$, baseline specification	0.740*** (0.096)	-0.691*** (0.143)	-1.984*** (0.592)
$\Delta O_s \times \text{Post}$, adding treated $\times$ year FE	0.751*** (0.100)	-0.747*** (0.147)	-1.016* (0.571)
Observations	240,448	240,448	240,448
Baseline controls $\times$ year FE	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes

Notes: The table probes whether shocks common to affected departments contribute to the dose-response estimates. The first row reports the baseline static specification. The second row adds a full set of treated-by-year fixed effects, which absorb any post-reform shock common to schools in Nariño and Valle del Cauca and identify the coefficient exclusively from variation in exposure among affected schools. All specifications include school and year fixed effects and interactions between year fixed effects and baseline municipality characteristics, on the balanced panel 2006–2018. Standard errors clustered at the school level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Own elaboration based on data from DANE.

D.4. Aggregate Effects: Synthetic Difference-in-Differences

We complement the dose-response estimates with aggregate estimates of the reform’s total effect using the synthetic difference-in-differences estimator described in Section 6 (Arkhangelsky et al., 2021), applied to the same balanced panel of rural public schools as our main specification. Whereas the dose-response design identifies the effect of the overlap channel from variation in exposure across schools, the SDID estimates capture the intent-to-treat effect of the reform on treated departments relative to a data-driven synthetic control. Appendix Figure D.5 presents the estimated dynamic paths for the three outcomes. We observe similar effects as with the dose-response strategy. Because treatment is assigned at the department level and only two departments are treated, formal inference at the level of treatment assignment is necessarily uninformative in this design, and school-level intervals understate uncertainty by treating schools as independent draws. We therefore read these aggregate estimates as descriptive, serving to gauge the magnitude of the reform’s total effect. The paper’s inferential claims rest on the dose-response design, where identification comes from variation in exposure across schools within the affected departments and standard errors can be clustered across roughly one hundred municipalities.

Appendix Figure D.5: Aggregate Effects of the Calendar Reform: Synthetic Difference-in-Differences

Notes: Each panel plots the estimated dynamic effects of the reform in each post-reform year, defined as the difference between treated schools and the synthetic control net of the time-weighted pre-reform mean difference, estimated on the balanced all-country panel of rural public schools used in the main specification. The synthetic control is constructed following the methodology described in Section 6. Vertical bars indicate 90%, 95%, and 99% confidence intervals based on bootstrapped standard errors clustered at the school level with 100 repetitions. Source: Own elaboration based on data from DANE.